UNIVERSITY OF CALIFORNIA, MERCED

Dark Matter: The Sub-galactic Frontier, A Computational Study of The Subhalos of Lensing Systems

A dissertation submitted in partial satisfaction of the

requirements for the degree

Doctor of Philosophy

in

Physics

by

Charles Gannon

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2026

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University of California, Merced

2026

DEDICATION

EPIGRAPH

It takes a hundred times to learn; a thousand times to understand.

—Zenyatta

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ACKNOWLEDGEMENTS

Firstly, I would like to thank all those who I collaborated with and those who provided support throughout my degree. I want to thank my advisor Anna Nierenberg for providing guidance and support throughout my PhD and pushing me to be a better scientist. In addition, I also would like to thank my committee members: Sarah Loebman (Chair), Anna Nierenberg, Lin Tian and Andrew Benson for their guidance throughout my degree. Furthermore, I want to extend a thank you to my brilliant collaborators: Andrew Benson, Xiaolong Du, Daniel Gilman, Ethan Nadler, Ryan Keeley and Andrew Robertson for their help and input. I would also like to thank my colleagues for the many productive discussions along the way to degree completion. Specifically, I would like to thank my lab members Dr. Oh and Dr. Perez as well as Jamie Quinn for bouncing around many ideas with me and providing their helpful insights. Next, I would like to an extend a thank you to Carnegie Institute of Science and Andrew Benson for providing office space for an extended stay at Carnegie Observatories and Andrew Benson for additional computational resources. Finally, I would like to extend a special thank you to Shelby Jackson for being my amazing partner and providing support in the final stretches of degree completion thanks for listening to many practice talks!

Next, I provide a list of discolsures:

This research was conducted using Pinnacles (NSF MRI, # 2019144) at the Cyberinfrastructure and Research Technologies (CIRT) at University of California, Merced.

This research was supported in part by grant NSF PHY-2309135 to the Kavli Institute for Theoretical Physics (KITP).

This research recieved support from the NSF through AST-2206315 "Collaborative Research: Measuring the physical properties of DM with strong gravitational lensing" and through JWST-GO program #2046 which was provided by NASA through a grant from the Space Telescope Science Institute, which is operated by the Associaion of Universities for Research in Astronomy, Inc., under NASA contract NAS 5-03127.

Generative AI Disclosure — Some research in this presentation was aided using

the help of Generative AI (LLM) coding agents and chats, specifically coding Claude models (Haiku, Sonnet, Opus, Fable across version 4.5-5.5). Generative AI was mostly used as a learning tool, coding assistant, literature review assistant. Some figures in this document were generated with the aid of these coding agents. Generative AI was not used in generating the text in this presentation.

Charles Gannon

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2021 **B.S. in Physics**, Idaho State University

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2. Incorporated principles of active learning and technological aids into discussion sections and labs

Publications

Charles Gannon, Anna Nierenberg, Andrew Benson, Ryan Keeley, Xiaolong Du, and Daniel Gilman. Dark matter substructure: A lensing perspective. *Physical Review D*, 112(2):023532, July 2025. doi: 10.1103/kk2n-q4ps.

D. Gilman, A. M. Nierenberg, T. Treu, C. Gannon, X. Du, H. Paugnat, S. Birrer, A. J. Benson, P. Mozumdar, K. C. Wong, D. Williams, R. E. Keeley, K. N. Abazajian, T. Anguita, V. N. Bennert, S. G. Djorgovski, S. H. Hoenig, A. Kusenko, M. Malkan, T. Morishita, V. Motta, L. A. Moustakas, W. Sheu, D. Sluse, D. Stern, and M. Stiavelli. JWST lensed quasar dark matter survey IV: Stringent warm dark matter constraints from the joint reconstruction of extended lensed arcs and quasar flux ratios. *arXiv e-prints*, art. arXiv:2511.07513, November 2025. doi: 10.48550/arXiv.2511.07513.

Ryan E. Keeley, A. M. Nierenberg, D. Gilman, C. Gannon, S. Birrer, T. Treu, A. J. Benson, X. Du, K. N. Abazajian, T. Anguita, V. N. Bennert, S. G. Djorgovski, K. K. Gupta, S. F. Hoenig, A. Kusenko, C. Lemon, M. Malkan, V. Motta, L. A. Moustakas, Maverick S. H. Oh, D. Sluse, D. Stern, and R. H. Wechsler. JWST lensed quasar dark matter survey - II. Strongest gravitational lensing limit on the dark matter free streaming length to date. *Monthly Notices of the Royal Astronomical Society*, 535(2):1652–1671, December 2024. doi: 10.1093/mnras/stae2458.

Xiaolong Du, Andrew Benson, Zhichao Carton Zeng, Tommaso Treu, Annika H. G. Peter, Charlie Mace, Fangzhou Jiang, Shengqi Yang, Charles Gannon, Daniel Gilman, Anna. M. Nierenberg, and Ethan O. Nadler. Tidal evolution of cored

and cuspy dark matter halos. *Physical Review D*, 110(2):023019, July 2024. doi: 10.1103/PhysRevD.110.023019.

A. M. Nierenberg, D. Gilman, T. Treu, X. Du, C. Gannon, H. Paugnat, S. Birrer, A. J. Benson, K. N. Abazajian, T. Anguita, S. G. Djorgovski, S. F. Hoenig, R. E. Keeley, A. Kusenko, H. R. Larsson, L. A. Moustakas, P. Mozumdar, W. Sheu, D. Sluse, D. Stern, D. Williams, and K. C. Wong. JWST Lensed Quasar Dark Matter Survey V: Measuring the minimum halo mass with strong gravitational lensing. *arXiv e-prints*, art. arXiv:2604.05237, April 2026. doi: 10.48550/arXiv.2604.05237.

Conferences & Presentations

2026 **American Astronomical Society, Winter 2026 Meeting** — Contributed Talk: "Accelerated Core Collapse of Highly Tidally Truncated SIDM Halos"

2024 – 2026 **UC Merced Cosmo Gal Research Symposium** — Contributed talks: "Dark Matter: A Lensing Perspective" and "T1DES in SIDM: A New Class of Self Interacting Dark Matter Subhalo Simulations"

2025 **American Physical Society, March Meeting 2025** — Poster: "Dark Matter: A Lensing Perspective"

2024 **Cosmic Signals of Dark Matter Physics: New Synergies** — Kavli Institute of Theoretical Physics, UC Santa Barbara — Contributed talk and poster: "Dark Matter: A Lensing Perspective"

2024 **KITP Workshop** — Kavli Institute of Theoretical Physics, UC Santa Barbara — Month long workshop on Dark Matter Theory, Simulation, and Analysis in the Era of Large Surveys.

Outreach and Awards

Outreach — Developed and taught sections of a week-long intensive for high school students covering simulation, modeling, and visualization of dark matter and gravitational lensing using Python.

Awards — Graduated from Idaho State University with Magna Cum Laude Distinction. University of California, Merced Physics Travel Fellowship.

ABSTRACT OF THE DISSERTATION

Dark Matter: The Sub-galactic Frontier, A Computational Study of The Subhalos of Lensing Systems

by

Charles Gannon

Doctor of Philosophy in Physics

University of California Merced, 2026

Dr. Sarah R. Loebman, Chair

Dark matter is thought to comprise the majority of the matter budget of the universe and serves a critical role in the history of our universe: baryons and dark matter collapse together, believed to be primordial fluctuations in the density field. Despite its important role in the evolution of our universe, its nature remains elusive. Large scale cosmological probes such as Planck, measurements of the spatial and mass functions of luminous galaxies the relative positions of dark matter and gas in the bullet cluster have provided evidence for cold and collisionless dark matter (CDM) on the largest scale of the universe.

Over the last few decades, new methods of probing smaller scale structures have emerged allowing CDM theories to be tested on the sub-galactic scale. Gravitational lensing is a particularly exciting avenue for probing this regime: it allows dark matter structure to be probed without the need for luminous tracers . Improved understanding of the predictions of dark matter models are needed to compliment the advance of observational probes to the sub-galactic scale. A significant challenge on this front is the simulation of subhalos, which are halos evolving within a tidal field of a larger galaxy. Subhalos present a unique challenge as they often undergo significant mass loss, drastically increasing the resolution necessary to resolve them in N-body simulations. In this work, we study numerical

predictions from analytic, Semi-analytic and N-body simulations for dark matter subhalos of galaxy lens scale halos in both CDM and alternate dark matter models. We develop informed priors on various population level quantities to support lensing studies of dark matter. This work is split into two main sections: first, we develop informed priors on the normalization of the halo mass function using semi-analytic models, second, we develop a new computationally efficient method for simulating SIDM subhalos in 1-D.

Many alternate dark matter models (such as warm dark matter or WDM) predict a suppression in the halo mass function, which is the differential number of halos per unit mass. In these models, there is a mass scale below which the number of halos is significantly reduced. Recent work has drawn attention attention to artificial disruption of subhalos in CDM N-body simulations, which may artificially lower predictions of the normalization of the subhalo mass function. To account for this theoretical uncertainty, lensing studies of dark matter have previously adopted wide priors on this normalization; however, this creates degeneracy when constraining properties of dark matter models such as any potential turnover in the halo mass function. To estimate current theoretical uncertainty in the subhalo mass function, we use the state-of-the-art semi-analytic `galacticus` model tuned to ultra high resolution N-body simulations that minimize the effects of artificial disruption. Using this model we develop informed priors on the projected halo mass function expected by lensing studies, develop models for the scaling of the halo mass function with redshift and halo mass and explore the tidal effects of a realistically evolving potential of a central galaxy. We find that a factor of ~ 2 accounts for theoretical uncertainties in the subhalos mass function and find that for current lensing studies the central galaxy has a currently negligible impact on the projected number density of substructure.

Recent inferences on the sub-galactic scale from direct gravitational imaging have found higher than predicted inner densities of dark matter halos. A explanations is a class of alternate dark matter models with non-negligible self-interaction on the sub-galactic scales, called self-interacting dark matter (SIDM); SIDM models with a sufficiently high cross section on the subgalactic scale can produce highly

dense halos due to gravothermal collapse. 3-D N-body simulations, which are traditionally the primary method of studying dark matter halos are computationally expensive especially in the core collapse regime; tidal mass loss amplifies this challenge by requiring a larger number of particles to resolve the halo initially, often requiring tens of thousands of core hours to properly resolve a single core collapse even in idealized simulations. In this work, we develop a pipeline to simulate the evolution of idealized SIDM subhalos in 1-D with realistic tidal evolution; by simulating subhalos evolution in 1-D, we are able to simulate SIDM physics much more computationally efficiently than the 3D case, resulting in over an order of magnitude speed up in some case enabling previously infeasible exploration of SIDM model space. Using our models, we explore the core collapse rates for a wide range for a wide range of SIDM cross sections for a realistic population of halos. We find that halos up to ~ 4 times as many halos collapse as subhalos when compared to identical at infall field halos if evaporation is negligible.

Chapter 1

Introduction

Dark matter is thought to comprise nearly 85% of the total mass of the universe Planck Collaboration [2020a]. Despite it's abundance, the nature of dark matter remains elusive. While large scale cosmological probes such as Planck Planck Collaboration [2020a] and surveys of the spatial and mass function distributions of galaxies White and Rees [1978], White and Frenk [1991], De Blok et al. [2008], Weinberg et al. [2015] and the relative positions of baryons and dark matter in the colliding bullet cluster Markevitch et al. [2004], Randall et al. [2008] have provided evidence for an empirical model of dark matter that is cold and collisionless (CDM) on the largest scale. Currently, innovative techniques using phenomena such as gravitational lensing [e.g. Vegetti et al., 2010, Gilman et al., 2019a, 2023, Minor et al., 2021] and stellar streams [e.g. Bonaca et al., 2019, Banik et al., 2021a] are being used to probe dark matter models on a new scale: the sub-galactic regime. The behavior of dark matter on the scale remains an area of active investigation and vigorous debate. Understanding this behavior of dark matter on this scale is not only interesting in it's own right, it may provide hints of the nature of dark matter matter. For example, certain particle physics models for sterile neutrinos predict structure formation distinct from CDM [Dodelson and Widrow, 1994]. Understanding the structure formation of our universe may provide guidance to dark matter experiments in the laboratory.

Gravitational lensing has emerged as a promising way to probe the sub-galactic regime, allowing low mass halos to be studied without the need for luminous tracers

and at cosmological distances. Two main approaches have been successful in constraining dark matter using gravitational lensing: direct gravitational to constrain individual objects [e.g. Vegetti et al., 2010, Minor et al., 2021] and anomalous flux ratio methods to constrain populations of objects [e.g. Gilman et al., 2019a, 2023]. In this work we mainly focus on flux ratios constraints using quadruply lensed quasars: in these systems, light from a quasar passes through a central galaxy and is gravitationally lensed, producing four observable images. The lensed light is lensed by the mass along it's line of site and due to the non-linear nature of the dependence of flux ratios on the second derivative of the potential, information on small scale dark matter substructure can be extracted from the from the flux ratios Gilman et al. [2019a]. To constrain dark matter models, forward model realizations of dark matter halos are generated across varying dark matter properties and a posterior is estimated based on the relative likelihood that a particular set of dark matter parameters match the observed flux ratios in a teqnique know as approximate Beysian computing. Several studies have used these methods to place some of the strongest constraints on Warm Dark Matter Gilman et al. [2019a], Nadler et al. [2021], Keeley et al. [2024] as well as the self-interaction cross sections of dark matter Gilman et al. [2023] and the primordial power spectrum Gilman et al. [2022]. In this work, study simulations and models of dark matter substructure of group mass ($\sim 10^{13} M_{\odot}$) to develop informed priors on dark matter models to support strong gravitational lensing studies. We split this in two sections: first we explore developing informed priors on the projected number density of subhalos within mass lensing halos, then we explore developing a new ultra fast simulation pipeline for simulating the evolution of dark matter halos in alternative dark matter models.

Perturbations in the density of dark matter gravitationally collapse into objects known as dark matter halos Press and Schechter [1974]; understanding how these perturbations grow and evolve is important for placing constrains on dark matter models. Dark matter halos merge hierarchically and remnants from previous mergers known as subhalos can remain gravitationally bound and continue to evolve within the host; these subhalos are effected by tidal forces from the host

and undergo mass loss due to tidal stripping and profile expansions due to tidal heating. N-body simulations suggest that subhalos account for 5–20% of the total number of halos near the lensed images in projection with the subhalo to isolated halo ratio depending strongly on redshift Xu et al. [2015], Despali et al. [2018]; although subhalos are subdominant, Gilman et al. [2024] showed that uncertainty in tidal stripping of models is a major source of uncertainty in strong gravitational lensing studies. Subhalos can be numerically challenging to study: subhalos can lose $> 99\%$ of their mass due to tidal stripping, leaving the subhalo prone to artificial noise in N-body simulations and possibly artificially disrupting, which would fictitiously lower the number of subhalos predicted by these simulations Errani and Navarro [2021]. To account for this, lensing studies have in the past, adopted wide priors on this normalization; however, this creates degeneracy when constraining properties of dark matter models such as any potential cutoff in the halo mass function. Many alternate dark matter models (such as warm dark matter or WDM) predict cutoff masses in differential number of halos per mass (the halo mass function); that is there is a mass scale below which the number of halos is significantly reduced. Therefore, implementing a narrower prior, informed by better understanding of tidal stripping could significantly increase the constraining of lensing studies.

In Chapter 2, we use a state-of-the-art semi-analytic `galacticus` Benson [2012] model tuned to ultra high resolution N-body to study populations of lensing like halos with the effects of artificial disruption minimized. We run a suite of `galacticus` models to study the substructure of lensing like halos; our spanning over an order of magnitude in halo mass and covering the redshift ranges gravitational lenses are expected at ($0.2 \leq z \leq 0.8$). Comparing our `galacticus` models to N-body simulations we estimate theoretical uncertainties and develop informed priors on the projected halo mass function expected by lensing studies. We study how this quantity can vary lens to lens; specifically we develop models for the scaling of the halo mass function with redshift and host halo using a combination of analytic and semi-analytic models. Finally, we investigate a potential shortcoming of many lensing specific dark matter simulations: the lack of baryonic potential.

We implement a realistic model in `galacticus` for a central massive elliptical using the Universe Machine Behroozi et al. [2019] model and study a suite of halos with this potential switched on. Driven by the difference between our Galacticus models and N-body simulations, we find that a factor of ~ 2 accounts for theoretical uncertainties in the subhalos mass function; this theoretical uncertainty is much than the nearly unbounded prior used by previous lensing studies. We provide a recalibration for the scaling of the projected subhalo mass function, and provide an explanation for this relation based on the geometry of the projection. Finally, we find that a central galactic potential causes a $\sim 15\%$ reduction in the normalization of the subhalo mass function of lens like systems, which is negligible for current lensing surveys.

Recent inferences on the sub-galactic from direct gravitational imaging the potential of lenses Vegetti et al. [2010], Minor et al. [2021], Tajalli et al. [2025] have provided evidence for anomalously dense dark matter halos compared to CDM predictions. Although these results have been hotly debated and solutions within the CDM framework have been proposed He et al. [2025], the intriguing possibility remains that these results hint at physics beyond CDM; possible explanations include a class of alternate dark matter models with non-negligible self-interaction on the sub-galactic scales, called self-interacting dark matter (SIDM). In these models, dark matter self-interaction facilitates the conduction of heat within the halo; due to the negative heat capacity of self-gravitating systems, runaway heat transfer can occur leading to the halo gravothermally collapsing, leading to extremely dense collapsed remnants at the end of their evolution Tulin and Yu [2018], Adhikari et al. [2025]. 3-D N-body simulations, which are traditionally the primary method of studying dark matter halos are computationally expensive especially in the core collapse regime; tidal mass loss amplifies this challenge by requiring a larger number of particles to resolve the halo initially, often requiring tens of thousands of core hours to properly resolve core collapse even in idealized simulations. Additionally, tidal forces from the host halo can alter the core collapse rates: in some cases decreasing the time for core collapse by up to or greater than a factor of 3 Nishikawa et al. [2020] or in some cases prevent it entirely Zeng et al. [2022],

depending on the specifics of the SIDM model. Understanding the effects of tidal forces on core collapse times in therefore critical to placing constraints on SIDM models.

In Chapter 3, we develop a pipeline to simulate the evolution of idealized SIDM subhalos in 1-D with realistic tidal evolution. By simulating subhalos evolution in 1-D, we are able to simulate SIDM physics much more computationally efficiently than the 3D case, resulting in over an order of magnitude speed up in some case enabling previously infeasible exploration of SIDM model space. To accomplish this, we augment the 1-D N-body code `NSphere` Kamionkowski et al. [2026], Kamionkowski and Sigurdson [2026] with a new method of simulating tidal physics in 1-D; we use a linear tidal field, with a strength dynamically calculated to reproduce the mass loss of a target halo. Our model is not only able accurately matches mass loss rates from N-body simulations, but also agrees well with internal properties of the halo such as the maximum circular velocity and the radius at which it occurs. Additionally, to further speed up our models we implement a model for adaptively calculating time steps based on scattering probabilities. We implement a pipeline to simulate SIDM evolution for a wide range for SIDM cross sections spanning an order of magnitude for a realistic population of 91 halos. Our longest running halos only take ~ 600 core hours to simulate, an order of magnitude speed up when compared to full 3-D simulations. We find that halos up to ~ 4 times as many halos collapse as subhalos when compared to identical at infall field halos if evaporation is negligible.

Chapter 2

Dark Matter Substructure: A Lensing Perspective

This chapter is reprinted from Charles Gannon, Anna Nierenberg, Andrew Benson, Ryan Keeley, Xiaolong Du, and Daniel Gilman. Dark matter substructure: A lensing perspective. *Physical Review D*, 112(2):023532, 2025. doi: 10.1103/kk2n-q4ps.

Abstract

The study of dark matter substructure through strong gravitational lensing has shown enormous promise in probing the properties of dark matter on sub-galactic scales. This approach has already been used to place strong constraints on a wide range of dark matter models including self-interacting dark matter, fuzzy dark matter and warm dark matter. One major uncertainty in these studies is the strength of tidal stripping experienced by subhalos. We study theoretical predictions for the statistical properties of subhalos in strong gravitational lenses using the semi-analytic galaxy formation toolkit: `galacticus`. We present a large suite of dark matter only `galacticus` models, spanning nearly two orders of magnitude in host halo mass (from Milky Way to group mass halos between redshifts from 0.2 to 0.8). Additionally, to address potential shortcomings of including only dark matter physics we include a smaller set of `galacticus` runs with the potential of

a central massive elliptical to complement our dark matter only suite of models. We place particular focus on quantities relevant to strong gravitational lensing; namely the projected number density of substructure near the Einstein radius as function of host stellar mass and redshift. In the innermost region in projection, we find that our `galacticus` models agree with N-body simulations within a factor of ~ 2 within the Einstein radius. We find that the addition of a central galaxy suppresses the projected number density of subhalos within the Einstein radius by around 15% relative to dark matter only simulations.

2.1 Introduction

Cold dark matter (CDM) provides an excellent description of the matter distribution on the largest scales For example, the CDM model has had tremendous success explaining the Cosmic Microwave Background [Planck Collaboration, 2020a] and the mass function and spatial distribution of luminous galaxies [e.g. White and Rees, 1978, White and Frenk, 1991, De Blok et al., 2008, Weinberg et al., 2015]. Recently, probes of dark matter have begun pushing the frontier to subgalactic scales without the need for luminous tracers. In this regime, stellar streams [e.g. Bonaca et al., 2019, Banik et al., 2021b,a, Bovy et al., 2017, Banik et al., 2018] promise to provide constraints within the Local Group. Outside the Local Group, strong gravitational lensing probes the properties of low mass halos [e.g. Vegetti et al., 2018, Minor et al., 2021, Powell et al., 2023, Mao and Schneider, 1998, Keeley et al., 2024, Dike et al., 2023, Gilman et al., 2019b,a, 2022, 2023].

One source of uncertainty in gravitational lensing analyses of dark matter is the normalization of the subhalo mass function (SHMF). Halos can be considered in two categories: 1) isolated (field) halos and 2) halos halo (subhalos). Both categories contribute to the lensing observables. While the properties of field halos can be robustly predicted for a given dark matter model, subhalos, undergo complex interactions with the host including tidal heating, tidal stripping and dynamical friction, which all occur within the evolving potential of the host halo. These physical effects map the subhalos at infall (the so-called *unevolved* population) to

the subhalos at a later time (referred to as the *evolved* population). An increase in tidal stripping results in a decrease in the normalization of the evolved SHMF. It is the *evolved* population of subhalos that have been affected by tidal interactions with the host that is observed during strong gravitational lensing measurements, whereas the unevolved population depends only on subhalo infall masses. Mass loss due to tidal stripping is spatially dependent Han et al. [2016] with some subhalos having evolved masses which are 1% or less of their infall masses Errani and Navarro [2021]. In the case of a subhalo with 1% of its original infall mass remaining, the potential lensing signal (percent change in image magnification) can be reduced by a factor of ~ 4 Du et al. [2025] compared to the signal that would be observed absent tidal interactions.

The subhalo to field halo ratio is strongly dependent on the lens and source redshift. N-body simulations suggest that subhalos account for 5–20% of the total number of halos near the lensed images in projection [Xu et al., 2015, Despali et al., 2018]. Although subhalos are subdominant in number, Gilman et al. [2024] showed that uncertainty in tidal stripping is a major source of degeneracy in strong gravitational lensing studies. Current forward modeling pipelines do not have variable models of tidal stripping strength; instead, the normalization of the unevolved SHMF ($\Sigma_{\rm sub}$) is allowed to vary. This serves as a proxy for varying tidal stripping strength. To study the impact of the $\Sigma_{\rm sub}$ prior on lensing inferences, Gilman et al. [2024] simulated 25 mock lenses with warm dark matter (WDM) with a half-mode-mass ($m_{\rm hm} = 10^{7.5}$ M$_\odot$) as a ground truth. Using a Gaussian prior on $\Sigma_{\rm sub}$ with width 0.2 dex lowered bound on the half-mode-mass by 2 dex relative to when a uniform prior with width of 1.5 dex was used. We aim to investigate whether a narrower prior on $\Sigma_{\rm sub}$ can be used given current theoretical uncertainties.

We highlight that caution must be exercised: a large body of work has drawn attention to artificial disruption present in N-body simulations [van den Bosch et al., 2017, Green and van den Bosch, 2019, Errani and Peñarrubia, 2019, Benson and Du, 2022]. Even in high resolution simulations, such as the Caterpillar [Griffen et al., 2016] suite of N-body simulations of Milky Way mass halos with a per particle mass of $\sim 10^4$ M$\odot$ artificial disruption can lead to a $10-20$% suppression of the

SHMF over the entire viral volume [Benson and Du, 2022]. Additionally, Benson & Du [Benson and Du, 2022] found a significant spatial dependence in the effects of artificial disruption, with the SHMF in the inner 2% of the virial radius spuriously suppressed by nearly a factor of 3. Due to the geometry of projection, subhalos at small radial displacement from the host are overrepresented in the populations of subhalos probed by lensing, making the effects of artificial disruption more pronounced when compared to the total subhalo population.

Due to their computational cost, zoom-in simulations are not conducive to studying population level statistics of subhalos over the large range of parameter space of host halo masses and redshift representative of lenses used to study dark matter. Analytic and semi-analytic models of subhalo interactions provide a path forward. These methods must account for the complex interactions between the subhalos and their host. Developing fast and accurate models of the impact of environment on subhalos remains an area of active study. In particular, models of tidal stripping have recently received a great deal of attention, with much recent work focused on understanding the tidal tracks of subhalo evolution [Errani and Peñarrubia, 2020, Errani and Navarro, 2021, Benson and Du, 2022, Du et al., 2024].

In light of the improved understanding of artificial disruption in N-Body simulations, this study revisits the predictions for the joint spatial and mass distribution of subhalos in CDM, with emphasis on quantities related to gravitational lensing. We use semi-analytic models as our primary mode of emulating dark matter subhalos. In particular, we use the `galacticus`[1] [Benson, 2012] galaxy formation toolkit. The `galacticus` galaxy formation toolkit is a modular and open source semi-analytic model of galaxy formation with an extensive library of both dark matter and baryonic physics.

We present a large suite of `galacticus` subhalo realizations, spanning nearly two decades in host halo mass (from Milky Way mass to group mass halos at redshifts 0.2–0.8). We use our suite of `galacticus` runs to study quantities relevant to strong gravitational lensing, with particular focus placed on the subhalo popu-

[1] https://github.com/galacticusorg/galacticus , we use revisions 723170fb1690257be9b0588c0a83c9e559a584ae and 0e3a5adea6e3b32b9eaaa3a6c4896bbe63fb0cb1

lation within a 20 kpc aperture. A 20 kpc aperture is chosen to be small enough to be comparable to the Einstein radius of a typical SLACS lens [Shu et al., 2017] ($\sim$ 1 arcsec corresponding to $\sim$ 8 kpc for a lens at redshift $z = 0.5$), but large enough to obtain reasonable statistics from our `galacticus` models. To complement our `galacticus` models, we include a comparison with the Symphony suite of N-body simulations and the Han et al. [2016] analytic model of the spatial and mass distribution of subhalos. Similar to previous works focusing on the subhalo populations of strong gravitational lenses, our main suite of models is limited to dark matter only physics. We make a preliminary study of baryonic effects by studying a smaller suite of models that include the potential of a central galaxy. We take an empirical approach to modeling the evolution of the central galaxy using the `UniverseMachine` [Behroozi et al., 2019] correlation between galaxy growth and dark matter halo assembly.

In Section 2.2, we discuss the parameters chosen for our `galacticus` models and the methods used in our analysis. Next, in Section 2.3 we present our results and provide discussion in Section 2.4. Finally, we summarize our results in Section 2.5. For this work, we assume cosmological parameters from the Planck Collaboration [Planck Collaboration, 2020a], $(H_0, \Omega_m, \Omega_\Lambda) = (67.36, 0.31530, 0.68470)$. A combination of `astropy` [Astropy Collaboration, 2013, 2018, 2022], `colossus` [Diemer, 2018] and `galacticus` [Benson, 2012] software packages are used for cosmological calculations. To calculate halo concentrations the Diemer and Joyce [2019] model is used. All quantities are reported in physical units unless otherwise specified.

2.2 Methods

In this section we discuss the methods used in this work. We discuss our semi-analytic models in Section 2.2.1. In Section 2.2.2 we provide an overview of theoretical expectations for the spatial distribution and mass function of subhalos.

2.2.1 Semi-Analytic Models

We utilize semi-analytic models as our primary mode of studying the population level statistics of dark matter substructure. Semi-analytic models have the advantage of speed when compared to N-body simulations, and are chosen to be the primary focus of this work to enable fast exploration of the population level statistics of the substructure of dark matter halos. In particular, we use `galacticus`, an open source and extensible galaxy formation framework. The `galacticus` toolkit is particularly well suited to the study of subhalo properties due to it's extensive library of subhalo physics and its recent calibration to high resolution idealized simulations provided in Du et al. [2024].

Here, we give a brief summary of the algorithms used to generate realizations of dark matter substructure. We use `galacticus` to generate merger trees and then subsequently evolve the trees forward in time using a set of analytic models for the evolution of subhalo density profiles and orbits (orbit initialization, dynamical friction, tidal stripping and tidal heating). In `galacticus`, dark matter merger trees are built using Monte Carlo algorithms presented in Cole et al. [2000a] and Parkinson et al. [2008]. In this algorithm, merger trees are evolved backwards in time, with branching rates calculated at each using extensions [Bond et al., 1991, Bower, 1991] to Press & Schechter [Press and Schechter, 1974] formalism. Merger tree nodes are then evolved forward in time. First, orbits are initialized using the potential of the host halo [Jiang et al., 2015]. After initialization, subhalo orbits are then evolved using models for dynamical friction [Chandrasekhar, 1943], tidal stripping [Zentner et al., 2005] and tidal heating [Gnedin et al., 1999].

Tidal Physics

As a subhalo evolves within its host halo, it gradually loses mass due to tidal stripping, i.e. particles are removed over time by the tidal forces exerted by the host. In addition, tidal shocks heat the subhalo, causing the density profile of the subhalo to expand and become less dense, causing further mass loss. For massive subhalos (subhalos with subhalo-to-host halo mass ratios close to 1), the trajectories of host particles are bent by the gravity of subhalos, creating an overdensity

behind the subhalo. This results in a net drag force known as dynamical friction, which causes the subhalo's orbital radius to decay and eventually leads to its merger with the host. As the subhalo moves closer to the host's center, it experiences increasingly stronger tidal forces, accelerating mass loss. In our work, we model subhalo evolution using the model described in Refs [Yang et al., 2020, Benson and Du, 2022, Du et al., 2024] within the `galacticus` framework.

We use subhalo physics tuned to high resolution idealized simulations provided by Du et al. [2024] in which subhalos are evolved in an analytic host potential. Compared to tradition cosmological zoom-in simulations, these simulations have the advantage of resolving subhalos at much higher resolution for less computational cost. For example, in the idealized simulations used by Du et al. [2024], subhalos start with $N \approx 10^7$ particles and are tracked until $N \approx 10^4$ particles remain. This is in contrast to high resolution cosmological simulations such as Caterpillar [Griffen et al., 2016], where the lowest mass subhalos are tracked to $N \approx 20$ particles. The high resolution minimizes the effects of numerical issues traditionally associated with N-body simulations, such as artificial disruption [van den Bosch and Ogiya, 2018]. Additionally, the challenges associated with halo finding in cosmological simulations is not present in idealized simulations. However, we note that our idealized simulations have several limitations; idealized simulations lack cosmological context and use simplified, spherically symmetric analytic profiles to describe the host halo and infalling subhalo.

Merger Radius

Probes of strong gravitational lensing are sensitive to subhalos appearing in projection near lensed images which appear near the Einstein radius. For a typical group mass lens at redshift $z = 0.5$, this aperture has a radius of ~ 1 arcsecond (~ 7 kpc) [Shu et al., 2017]. Due to effects of projection, strong lensing probes are especially sensitive to subhalos at small spatial separation to the host. The `galacticus` galaxy formation model automatically destroys subhalos within a minimum radius. We select the minimum radius to simulate subhalos by running convergence tests. We find convergence for the spatial distribution at 10 kpc for a

merging radius of 0.01 $r_{\rm v}$. For a $10^{13.5}{\rm M}_\odot$ mass halo at redshift, z = 0.2, 0.01 $r_{\rm v}$ corresponds to a radial separation from the center of the host of 0.7 kpc. Unless otherwise noted, when considering projected quantities, we include subhalos within an annulus with an inner and outer radius of 10 and 20 kpc respectively. This ensures that the subhalo population within 10 kpc, which may be incomplete, is excluded. We run our models with a tree (infall mass) floor of 8×10^7 ${\rm M}_\odot$. To ensure the completeness of the subhalo population, we track subhalos to arbitrarily low bound mass.

Host Halo Properties

We model across a broad range of masses and redshifts to capture the evolution of the projected number density and to facilitate comparison with observational works as well as other simulations. Our suite of `galacticus` models consists of a 10 by 10 grid of `galacticus` outputs spanning the range of host halo masses $12.0 \leq \log_{10}\left(M_{\rm h}/{\rm M}_\odot\right) \leq 13.5$ and redshifts $0.2 \leq z \leq 0.8$ with 224 host halos per grid point.

Central Galaxy Properties

In addition to our dark matter only models, we include a smaller suite of `galacticus` models spanning $13.0 \leq \log_{10}(M_{\rm h}/{\rm M}_\odot) \leq 13.5$ with the inclusion of the potential of a central massive elliptical. Along with the increased tidal heating and stripping due to the potential of the central galaxy, we include a model of baryonic contraction due to the central potential [Gnedin et al., 2004]. The morphology of the central massive elliptical is modeled using a Hernquist [Hernquist, 1990] profile. We take care to accurately model the evolution of the stellar mass and scale radius of the central galaxy; the evolution of the central galaxy is modeled using the stellar mass to halo mass relation of Behroozi et al. [2019], Equation J3. For our work, we use the best fit parameters provided in the first row of table J1. For simplicity, we do not include intrinsic scatter in the stellar mass to halo mass relation. We use empirical power law fits provided by Shen et al. [2003] for the stellar mass to radius relationship for early type galaxies

in the Sloan Digital Sky Survey [York et al., 2000] to evolve the Hernquist radius of the central galaxy. Similarly, we do not include scatter in this relationship. We provide further discussion of the evolution of the central galaxy in Appendix 2.A.

2.2.2 Analytic Models

Here, we discuss theoretical expectations for the subhalo population. We place particular emphasis on the spatial and mass distribution of subhalos. Two cases of the spatial-mass distribution can be considered, using two definitions of subhalo mass: the mass at infall and the gravitationally bound mass of the subhalo at the time of observation. The former is known as the "unevolved" distribution and the latter the "evolved" distribution. By definition, no mass loss due to tidal stripping is considered in the unevolved distribution, while the evolved distribution includes tidal mass loss. Due to the extreme mass loss undergone by some subhalos, the unevolved distribution is not an observable quantity. Instead, the unevolved distribution provides a powerful tool to study the properties of subhalos, allowing the complicated physics of tidal stripping to be separated from orbital physics. A useful picture is that the evolved distribution can be approximately thought of as being derived from the unevolved distribution and a model of tidal mass loss [Han et al., 2016, Du et al., 2025]. However, we note that this picture is not exact. For example, physics of tidal stripping cannot be completely separated from orbital physics as tidal mass loss can affect dynamical friction (`galacticus` uses the bound mass in dynamical friction calculations).

We compare the predictions from N-body simulations and semi-analytic models with the analytic prescription presented by Han et al. [2016] . Here, we describe the Han et al. [2016] model and discuss some of its physical assumptions. Han et al. [2016] first assume that the unevolved spatial-mass distribution of subhalos is separable (the spatial and mass function can be separated into a mass function and spatial distribution with the mass function only dependent on the subhalo's mass and the spatial distribution only dependent on the subhalo's radial separation from the host). Then Han et al. [2016] combine this assumption with the assumption that the unevolved spatial distribution of subhalos traces the density profile of the

host to arrive at the unevolved spatial and mass distribution:

$$\frac{\mathrm{d}^2N}{\mathrm{d}V\mathrm{d}m} \propto \tilde{\rho}(r)\frac{\mathrm{d}N}{\mathrm{d}m}(m), \tag{2.1}$$

where $\frac{\mathrm{d}^2N}{\mathrm{d}V\mathrm{d}m}$ is the unevolved subhalo spatial and mass function, $\tilde{\rho}$ is the normalized density profile of the host, $\mathrm{d}N/\mathrm{d}m$ is the unevolved mass function and m is the unevolved (infall) mass of the subhalo. The unevolved mass function is then approximated as a power law

$$\frac{dN}{dm} \propto m^{\alpha}, \tag{2.2}$$

with $\alpha \sim -2$ being the logarithmic slope of the subhalo mass function. Next, Han et al. [2016] assume that the gravitationally bound (evolved) mass of a subhalo can be related to the unevolved mass of the subhalo on average using a transfer function dependent only on the distance from center of the host

$$\frac{m_b}{m} = T(r), \tag{2.3}$$

where $T(r)$ is the spatially dependent transfer function and m_b is the evolved mass of the subhalo. In their analysis, Han et al. [2016] take $T(r)$ to be a power law

$$T(r) \propto \left(\frac{r}{r_{\mathrm{v}}}\right)^{\beta}, \tag{2.4}$$

where $\beta \sim 1$ is a free parameter.

Plugging Equations 2.2 and 2.3 into Eq. 2.1 gives the evolved spatial and mass function

$$\frac{\mathrm{d}^2N}{\mathrm{d}V\mathrm{d}m_{\mathrm{b}}} \propto \hat{T}(r)\tilde{\rho}(r)m_b^{\alpha}, \tag{2.5}$$

where $\frac{\mathrm{d}^2N}{\mathrm{d}V\mathrm{d}m_{\mathrm{b}}}$ is the evolved spatial and mass function and $\hat{T}(r) = (T(r))^{-(\alpha+1)}$ is substituted for brevity. We note that for $\alpha \sim -2$, $\hat{T}(r) \approx T(r)$. Furthermore, substituting the form of $T(r)$ used by Han et al. [2016] (Eq. 2.4) gives

$$\frac{\mathrm{d}^2N}{\mathrm{d}V\mathrm{d}m_{\mathrm{b}}} \propto r^{\gamma}\tilde{\rho}(r)m_b^{\alpha}, \tag{2.6}$$

where $\gamma = -\beta(\alpha+1) \sim 1$. Han et al. [2016] fit γ to various N-body simulations finding $\gamma = 0.95$ for Milky Way mass Aquarius halos [Springel et al., 2008] and $\gamma = 1.33$ for group mass halos in the Phoenix suite [Fiacconi et al., 2016], indicating a dependence on host halo mass.

To analyze our `galacticus` results, we fit the Han et al. [2016] models to the `galacticus` predictions for the spatial and mass function in Section 2.3. To provide an additional point of comparison to our `galacticus` results in the context of gravitational lensing, we spatially project the Han et al. [2016] into 2d. An immediate consequence of the separability assumed by the Han et al. [2016] model is a prediction that the logarithmic slope of the SHMF should be identical between the projected and unprojected distributions. Additionally, we study the predictions of the model for the scaling of the projected subhalo mass function (PSHMF) within the Einstein radius, an important quantity for gravitational lensing. We estimate the scaling of the projected spatial and mass function as a function of host halo mass utilizing the host halo mass scaling relations discussed in Han et al. [2016]. However, redshift dependent scaling is not considered by Han et al. [2016] , who derived their model for a single snapshot in time at $z = 0$. Typical lensing galaxies are found at redshifts $0.2 \lesssim z \lesssim 0.8$, so understanding the scaling of the SHMF with redshift is essential. Therefore, we discuss extending the Han et al. [2016] model over these range of redshifts in the following sections.

2.3 Results

Here, we analyze our suite of `galacticus` models. In Section 2.3.1 we discuss the spatial distribution predicted by `galacticus`. Next, in Section 2.3.2 we calculate the normalization of the PSHMF of our dark matter only (DMO) simulations, and repeat the calculations for our models with the potential of a central massive elliptical in Section 2.3.3. Next, in Section 2.3.4 we predict the scaling of the PSHMF as a function of host halo mass and redshift using our dark matter simulations. Finally, we discuss the impact of a central galactic potential on the scaling of the PSHMF.

2.3.1 Spatial Distribution

Figure 2.1 shows the spatial distribution predicted by `galacticus` for a $10^{13}\mathrm{M}_{\odot}$ halo at $z = 0.5$, with both the evolved and unevolved distributions plotted. In addi-

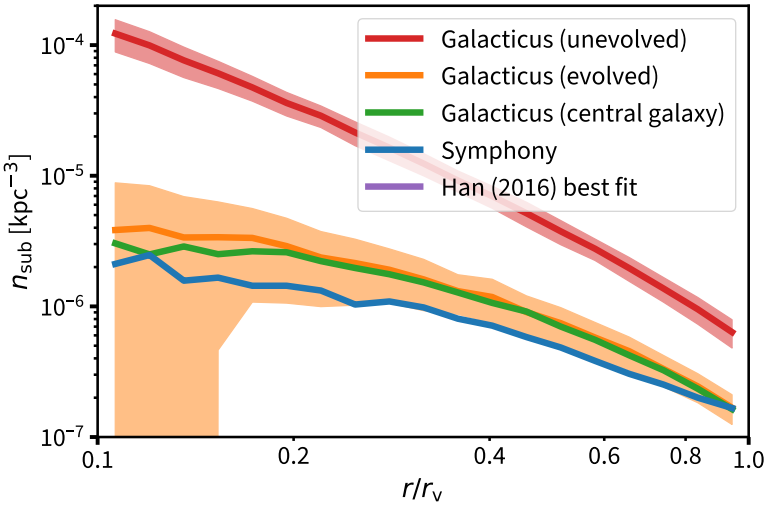


Figure 2.1: The differential spatial distributions ($n_{\rm sub}$) from our `galacticus` models and Symphony for subhalos with mass $9 < \log_{10}(m_b/{\rm M}_\odot) < 10$ plotted as a function of the virial radius fraction ($r/r_{\rm v}$). Results are for halos with mass $M_{\rm h} = 10^{13}{\rm M}_\odot$ at redshift $z = 0.5$ In red: the unevolved spatial distribution from our `galacticus` models, in orange: the evolved spatial distribution for `galacticus` and in green with a central galaxy. The `galacticus` and Symphony spatial distributions shown are a result of averaging over the spatial distributions of each individual halo in the suite at each bin over all halos in the suite. The halo to halo scatter is calculated in a similar way; taking the standard deviation of the values of the spatial distributions of each of the host halos calculated at each radial bin. The shaded regions show the 1σ halo to halo scatter for the `galacticus` evolved and unevolved case, scatter for the other distributions is not shown for visual clarity. The best fit of the Han (2016) model to the `galacticus` DMO unevolved spatial distribution is shown in purple. In blue: the spatial distribution for Symphony. We find there is no significant difference in the spatial distribution of subhalos when running models with or without a central galaxy.

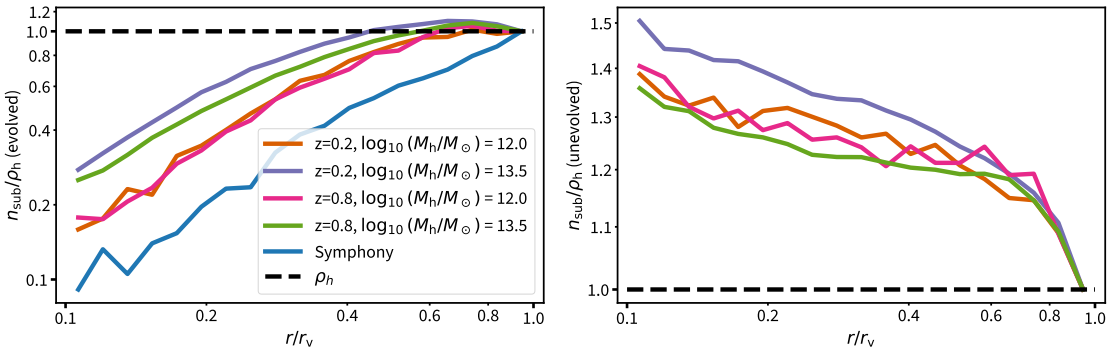


Figure 2.2: The ratio of the spatial distribution of subhalos ($n_{\rm sub}$) to the host's density profile ($\rho_{\rm h}$) for subhalos with $8 < \log_{10}(m_b/{\rm M}_\odot) < 9$ normalized to 1 at $r = r_{\rm v}$ for the evolved distribution (left) and $8 < \log_{10}(m/{\rm M}_\odot) < 9$ unevolved distribution (right). A horizontal line of 1 (black dotted) shows the host halos density profile. Results are plotted for `galacticus` over a range of host halo masses and redshifts and for Symphony for a single mass and redshift ($\log_{10}(M_{\rm h}/{\rm M}_\odot) = 13, z = 0.5$). The relative decrease towards the center in the evolved case is a result of tidal stripping (see Section 2.2.2), while the relative increase in the unevolved case is likely a result of dynamical friction (see Section 2.4.1). Over our range of redshifts explored by our `galacticus` simulations, the slope of the evolved spatial distribution $n_{\rm sub}/\rho_{\rm h}$ has a mild dependence on halo mass, with almost no dependence on redshift. For the unevolved case, there appears to be no significant dependence one either redshift or host halo mass.

Simulation	$\log_{10}(M_{\rm h}/{\rm M}_\odot)$	z	γ
`galacticus`	12.0	0.2	1.01
	12.0	0.8	1.06
	13.0	0.5	0.82
	13.5	0.2	0.67
	13.5	0.8	0.75
Aquarius	12.0	0.0	0.95
Phoenix	14.8	0.0	1.33

Table 2.1: Table of best fits to Han et al. [2016] model of the spatial distribution (see Eq. 2.5). We include fits spanning the range of halo masses and redshifts included in our suite of `galacticus` models, as well as at the redshift and host halo mass, $M_h = 10^{13}{\rm M}_\odot$ and redshift, $z = 0.5$ where $\Sigma_{\rm sub}$ is defined. For comparison, we include fits to Aquarius and Phoenix halos provided by Han et al. [2016] .

tion to the `galacticus` predictions, results are shown for Symphony [Nadler et al., 2023] and Han et al. [2016] model of the spatial distribution. We plot the ratio of the radial distribution of subhalos to the host's dark matter density in Figure 2.2. For a $10^{13}{\rm M}_\odot$ halo at $z = 0.5$ we find the best fit to the `galacticus` spatial distribution with a value of $\gamma = 0.98$ (shown in Table 2.1). Additionally, fits provided by Han et al. [2016] to Aquarius [Springel et al., 2008] and Phoenix [Fiacconi et al., 2016] are included for comparison in Table 2.1.

The fits to the $10^{12.0}$ ${\rm M}_\odot$ `galacticus` halos agree well with fits to Aquarius halos ($\approx 10^{12.0}$ ${\rm M}_\odot$). We find a trend of decreasing γ with host halo mass in our suite of `galacticus` models. This is opposite of the trend of increasing γ with halo mass found by Han et al. [2016] when comparing the Aquarius and Phoenix simulations ($\approx 7 \times 10^{14}{\rm M}_\odot$). Over the range of redshifts probed $0.2 \leq z \leq 0.8$, and at a fixed halo mass we find that γ is nearly constant. Further discussion of the Han et al. [2016] model is provided in Section 2.4.1.

Figure 2.3 shows the mean and standard deviation of the projected spatial distributions predicted by `galacticus` for subhalos in the mass range $8 < \log_{10}(m/{\rm M}_\odot) < 9$ for a 10^{13} ${\rm M}_\odot$ halo at $z = 0.5$ with projected radii $r_{\rm 2d} < 20$ kpc. The mean and standard deviation are shown for each 76 kpc^2 radial bin, combining an aggregate of 3 projected spatial distributions (in the xy, yz, zy planes) per host halo.

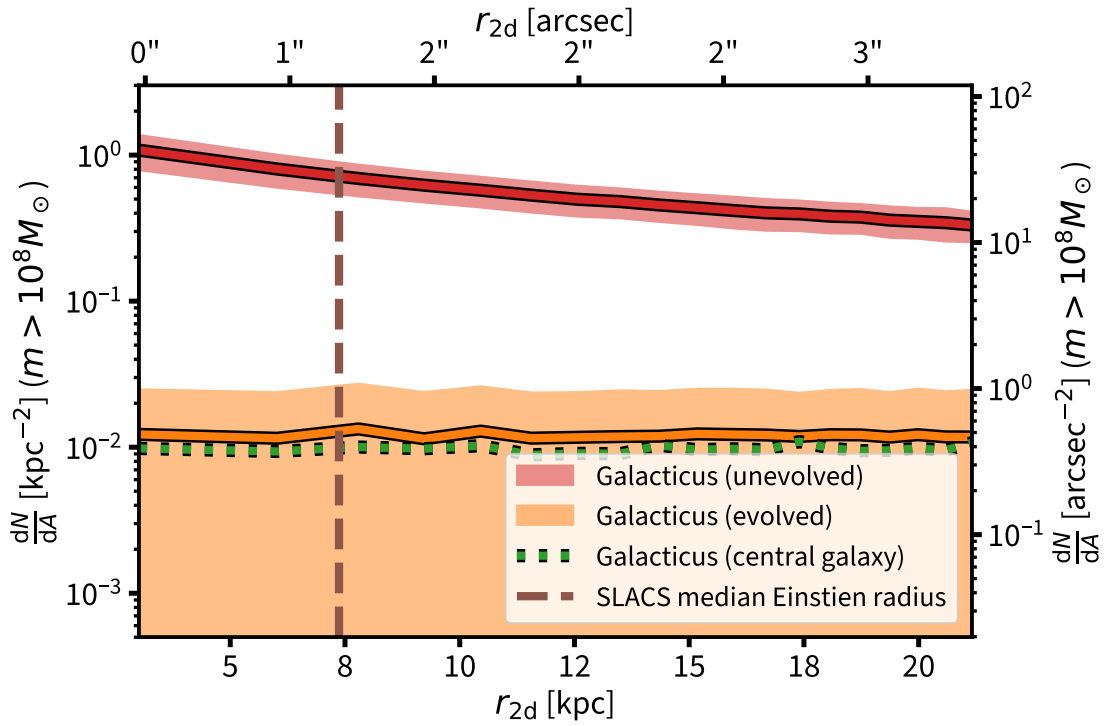


Figure 2.3: The projected number density within the inner 20 kpc for both the unevolved (upper red), evolved with central galaxy (green dotted) and evolved (lower orange) subhalo distributions. Predictions from `galacticus` are shown for subhalos with mass $> 10^8 M_\odot$ in a $10^{13} M_\odot$ halo, with scatter shown for 76 kpc^2 bins. In brown (dotted) the Einstein radius for a typical SLACS lens is shown. The evolved number density is constant within the innermost region of the halo for both DMO and central galaxy models. The solid line shows the mean, while the shaded region shows 1σ halo to halo scatter (see 2.3.1 for more details). Scatter is not shown for the case with a central galaxy for visual clarity. The projected spatial distribution with the inner 20 kpc is nearly independent of projected distance. Additionally, for our group scale simulations, we find that the addition of a central massive elliptical has a negligible impact on the projected number density of subhalos.

For reference, we show the median Einstein radius from the sample of SLACS lenses presented by Shu et al. [2017]. We find that the projected spatial distribution is nearly constant in the innermost region of the host, which is consistent with the findings of Xu et al. [2015]. We find that this holds true for both our DMO models and models with a central galaxy.

2.3.2 Projected Subhalo Mass Function Normalization: Dark Matter Only

Of particular interest to lensing studies is the integrated number of subhalos within an aperture near the Einstein radius as a function of the properties of the host halo. To measure this, we use the convention of Gilman et al. [2019a] to define the average density of the unevolved projected subhalo mass function (PSHMF) within the Einstein radius as:

$$\frac{\mathrm{d}^2N}{\mathrm{d}m\mathrm{d}A} = \frac{\Sigma_{\mathrm{sub}}}{m_0}\left(\frac{m}{m_0}\right)^{\alpha} F(M_{halo}, z), \tag{2.7}$$

where $\frac{\mathrm{d}^2N}{\mathrm{d}m\mathrm{d}A}$ is a measure of the projected subhalo number density evaluated at the Einstein radius, m is the mass of a subhalo at infall, m_0 is a pivot mass, taken by convention to be $10^8\mathrm{M}_{\odot}$, and the function $F(M_{halo}, z)$ describes the dependence on host halo mass and redshift. For convenience, we refer to $\frac{d^2N}{dmdA}$ as the projected subhalo mass function (PSHMF) (instead of "projected subhalo mass density function"). We note that Eq. 2.7 can be derived by projecting Eq. 2.1. The scaling term $F(M_h, z)$ is a result of the dependence of the total number of subhalos on halo mass and redshift as well as the dependence of the spatial distribution of subhalos on both halo mass and redshift (for the unevolved case, Han et al. [2016] models the spatial distribution of subhalos as following the density profile of the host). Factoring out all dependence on redshift and host halo mass into the scaling function $F(M_{halo}, z)$ allows a single PSHMF normalization (Σ_{sub}) to be measured for all host halos regardless of host halo mass or redshift.

To provide further context for Eq. 2.7, we can define the evolved PSHMF:

$$\frac{\mathrm{d}^2N}{\mathrm{d}m_{\mathrm{b}}\mathrm{d}A} = \frac{f_s \cdot \Sigma_{\mathrm{sub}}}{m_0}\left(\frac{m_b}{m_0}\right)^{\alpha_{\mathrm{b}}} F_b(M_{halo}, z), \tag{2.8}$$

where f_s is the reduction in normalization from the unevolved case, $\alpha_{\rm b}$ is the logarithmic slope of the evolved SHMF and $F_b(M_h, z)$ described the halo mass / redshift scaling of the evolved PSHMF. Eq. 2.8 can be derived using a similar procedure to that used by Han et al. [2016] : define an average ratio of subhalo bound mass to infall mass for those subhalos within the Einstein radius $m/m_b = f_m$ and project Eq. 2.5. This derivation yields $\alpha_b = \alpha$ and $f_s = f_m^{-(\alpha+1)}$, which for $\alpha \sim -2$ can be approximated $f_s \approx f_m$.

We include best fits for Eq. 2.7 and 2.8 in Table 2.2. As shown in Table 2.2, we find subhalos of a group scale host that appear near the Einstein radius in projection on average retain $\sim 2\%$ of their infall mass after tidal stripping. Similar to the unevolved case, the scaling term ($F_b(M_h, z)$) results from the dependence of the total subhalo count and the spatial distribution of subhalos on both halo mass and redshift. We note the spatial distributions differ in the evolved and unevolved case; therefore we should expect $F_b(M_h, z) \neq F(M_h, z)$.

Eq. 2.7 and 2.8 can be derived by spatially projecting the Han et al. [2016] models of spatial and mass distribution into 2d. The Han et al. [2016] model predicts that the logarithmic slopes of the evolved and unevolved mass functions should be identical, as well as that the logarithmic slopes of the total and projected mass functions should be identical. We check these assumptions against our model results in Table 2.4. Furthermore, we discuss why this assumption may be inaccurate (at least inaccurate for the $> 10^8 M_\odot$ halos considered here) due to mass segregation from dynamical friction in the unevolved distribution in section Section 2.4.1. For simplicity, in the remainder of this work we assume $\alpha = \alpha_{\rm b} = -1.97$ to remain consisistent with the `galacticus` results for the evolved distribution (see Table 2.4).

All dependence on halo mass and redshift in Eq. 2.7 and Eq. 2.8 is captured in F and F_b respectively, making $\Sigma_{\rm sub}$ and $f_{\rm s}$ independent of halo mass and redshift. Fits of the Han et al. [2016] model to the `galacticus` spatial distribution are given in Table 2.1. Best fits for $\Sigma_{\rm sub}$, $f_{\rm s}$ and α for subhalos within an annulus with inner and outer radii of 10 kpc and 20 kpc are given in Table 2.2. We choose a $10 - 20$ kpc annulus to exclude the innermost subhalo population which may be subject to

destruction due to satellite merging implemented in `galacticus`, as is discussed in Section 2.2.1.

Note that because the evolved projected spatial distribution is nearly independent of radius near the center of the host, the normalization of the evolved mass (density) function ($f_{\rm s} \cdot \Sigma_{\rm sub}$) will be nearly independent of the choice of aperture radius. We compare $f_{\rm s} \cdot \Sigma_{\rm sub}$ to the PonosV [Fiacconi et al., 2016] and PonosQ [Fiacconi et al., 2016] simulations in Table 2.3.

2.3.3 Projected Subhalo Mass Function Normalization: Impact of Central Galaxy

Figure 2.1 shows the impact of the galaxy on the spatial distribution, while Appendix 2.A gives more information on the evolution of the central galaxy in our `galacticus` models. Our `galacticus` models predict the central galaxy has a minimal impact on the SHMF, as shown in Figure 2.5. Within the inner 20 kpc in projection, both the DMO and galactic potential results have no dependence on radius as shown in Figure 2.3. For low mass halos, our models predict the central galaxy introduces a mass independent rescaling of the SHMF, with no change in slope. The rescaling of the SHMF is dependent on the distance from the host, with the SHMF in the inner region more heavily suppressed when compared to the DMO predictions. For a $10^{13}{\rm M}_{\odot}$ halo, the SHMF is suppressed at the $< 5\%$ level over the entire virial volume when compared to DMO predictions. In the inner $10-20$ kpc annulus the suppression increases to $\sim 15\%$. We find the impact of the central galaxy is minor when compared to theoretical uncertainties in the SHMF, the difference in normalization between `galacticus` and Symphony is greater than the difference between `galacticus` with and without a central galaxy.

2.3.4 Projected Subhalo Mass Function Scaling: Dark Matter Only

To parameterize F and F_b we follow the convention of Gilman et al. [2019b] and use the power law expansion

$$F(M_{halo}, z) = \left(\frac{M_{halo}}{10^{13}\mathrm{M}_\odot}\right)^{k_1} (z + 0.5)^{k_2} \tag{2.9}$$

defined to be normalized to 1 at the host halo mass and redshift of a typical lens ($M_\mathrm{h} = 10^{13}\mathrm{M}_\odot$ and $z = 0.5$). A powerlaw is a natural aproximation due to the nearly scale-free evolution expected in CDM.

We fit k_1 and k_2 using our suite of `galacticus` models. We plot the scaling relation in Figure 2.4, and give scaling coefficients in Table 2.4. We find best fit values of $k_1 = 0.55$ (halo mass scaling coefficient) and $k_2 = 0.37$ (redshift scaling coefficient) for the unevolved distribution and $k_1 = 0.37$ and $k_2 = 1.05$ for the evolved distribution.

The scaling of the PSHMF depends on both the spatial and mass distributions. To model the PSHMF scaling analytically, we project the Han et al. [2016] model. To scale the normalization of the total SHMF (over the entire virial volume) as a function of host halo mass and redshift we use the relations provided by Van den Bosch et al. [2005] for the evolved case and Gao et al. [2004] for the unevolved case. We project Eq. 2.1 and 2.5 according to the projection equation:

$$\frac{\mathrm{d}^2 N}{\mathrm{d}A\mathrm{d}m}(r_{2d}) = 2\int_{r_{2d}}^{r_\mathrm{v}} \frac{u}{\sqrt{u^2 + r_{2d}^2}} \frac{\mathrm{d}^2 N}{\mathrm{d}V\mathrm{d}m}(u)\mathrm{d}u, \tag{2.10}$$

assuming a spherically symmetric distribution of subhalos, so that the number density of subhalos depends only on the distance from the center of the host, r. Using the formula provided in Van den Bosch et al. [2005], the amplitude of the total unevolved SHMF scales as a function of mass and redshift according to the relation:

$$\frac{\mathrm{d}N}{\mathrm{d}(m/M_\mathrm{h})} = C_u \left(\frac{m}{M_\mathrm{h}}\right)^\alpha, \tag{2.11}$$

where m is the mass of the subhalo (at the time of accretion), M_h is the mass of the host halo, α is the logarithmic slope and C_u is the normalization of the unevolved

SHMF. For the evolved SHMF, we scale the amplitude using the empirical formula provided by Gao et al. [2004]:

$$\frac{\mathrm{d}N}{\mathrm{d}m_{\mathrm{b}}} = C_{\mathrm{e}} M_{\mathrm{h}} f(m_{\mathrm{b}}, z) m_{\mathrm{b}}^{\alpha}, \tag{2.12}$$

where C_{e} is the normalization of the evolved SHMF, m_{b} is the bound mass of the subhalo and $f(m_{\mathrm{b}}, z)$ abundance of halos with mass m_{b} at redshift z per unit mass in the universe. We compute $f(M_{\mathrm{h}}, z)$ using the Sheth & Tormen [Sheth and Tormen, 1999] mass function.

To calculate the projected scaling relation, we numerically integrate Eq. 2.5. We first fix γ as constant for all host halo masses, and calculate the projected scaling relations in halo mass and redshift. The range $0.8 \leq \gamma \leq 2.0$ is chosen to represent a reasonable range of possible values of γ, since $\gamma \approx 1$ is expected. The limit as γ goes to 0 gives a NFW profile (which is the same as the unevolved case). For $\gamma > 3$, the spatial distribution is increasing as a function of r for all values of r, which is nonphysical. In addition to a static value of γ, we consider an additional case where γ varies as a function of halo mass ("γ interp"). In this case, we linearly interpolate γ as a function of halo mass, using fits to the spatial distribution predicted by `galacticus` (see Table 2.1). We summarize our scaling models by providing fits to Eq. 2.9 in Table 2.2. We also use these scaling relations in Table 2.3 to compare our measured values to results from the Aquarius [Springel et al., 2008] and Phoenix [Gao et al., 2012] simulations as well as the Milky Way prior used Nadler et al. [2021].

2.3.5 Projected Subhalo Mass Function Scaling: Impact of central galaxy

We find the central galaxy has only a minor impact on the scaling of the amplitude of the PSHMF in redshift and host halo mass (see Table 2.4). For more massive halos ($M_h > 10^{13}\mathrm{M}_{\odot}$), the difference when compared to the Dark Matter Only (DMO) model decreases, which is expected due to the decreasing baryon fraction of the central galaxy (see Figure 2.6). Unless otherwise stated, for the remainder of this work we use the DMO scaling relation.

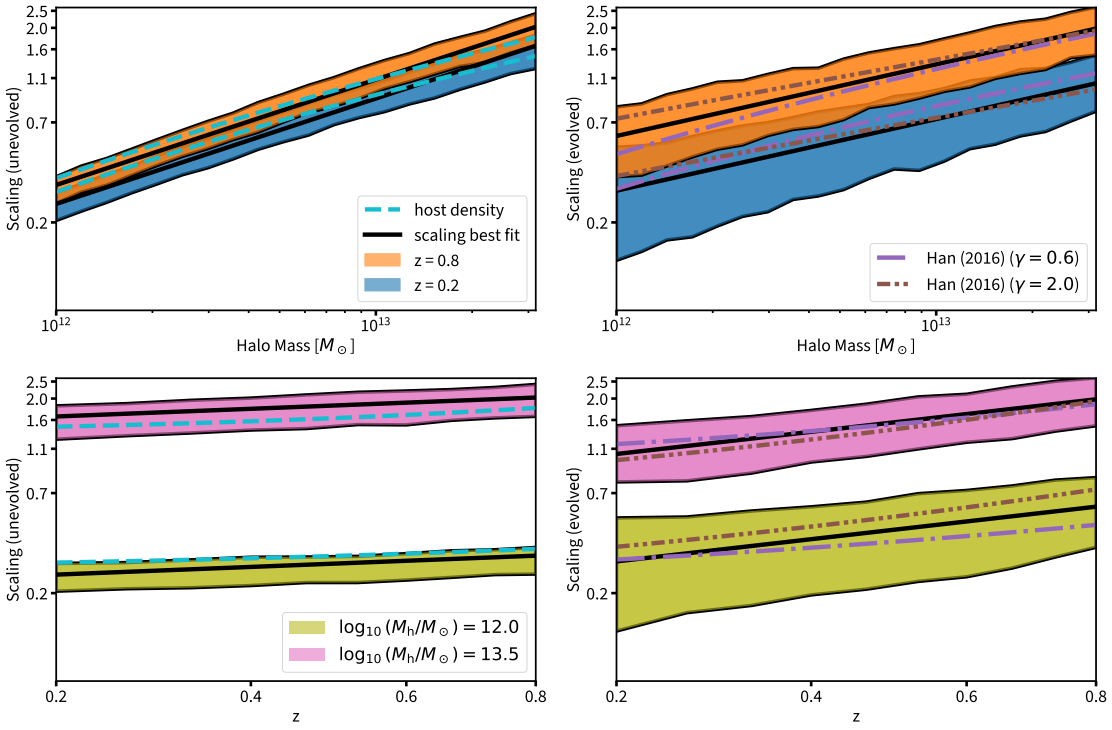


Figure 2.4: The scaling shows the ratio of number of subhalos with mass $8 < \log_{10} m < 9$ projected within an annulus with an inner and outer radius of 10 and 20 kpc respectively to the number of these subhalos in a $10^{13} M_\odot$ host halo at $z = 0.5$ as a function of host halo mass and redshift. The scaling value and its scatter are calculated by using the mean and standard deviation over all host halos. The left panels show the unevolved subhalo scaling, while the right panel shows the scaling of the evolved distribution. Upper panels show the halo mass scaling relation for host halos at redshift $z = 0.2$ (blue) and $z = 0.8$ (orange). Lower panels show the redshift scaling relation for a host halo mass of $10^{12.0}\mathrm{M}_\odot$ (olive) and $10^{13.5}\mathrm{M}_\odot$ (pink). Shaded regions show the 1σ halo to halo scatter.d The unevolved scaling is compared to the scaling of the host's density profile (adjusted to the definition of halo mass used here) in blue (dashed). The scaling relations derived from the Han et al. [2016] analytic number density distribution (Eq. 2.1 and 2.5) are shown for $\gamma = 0.6$ (purple, dot dashed) and $\gamma = 2.0$ (brown, dot dashed). These values of γ span a range of physically plausible values.

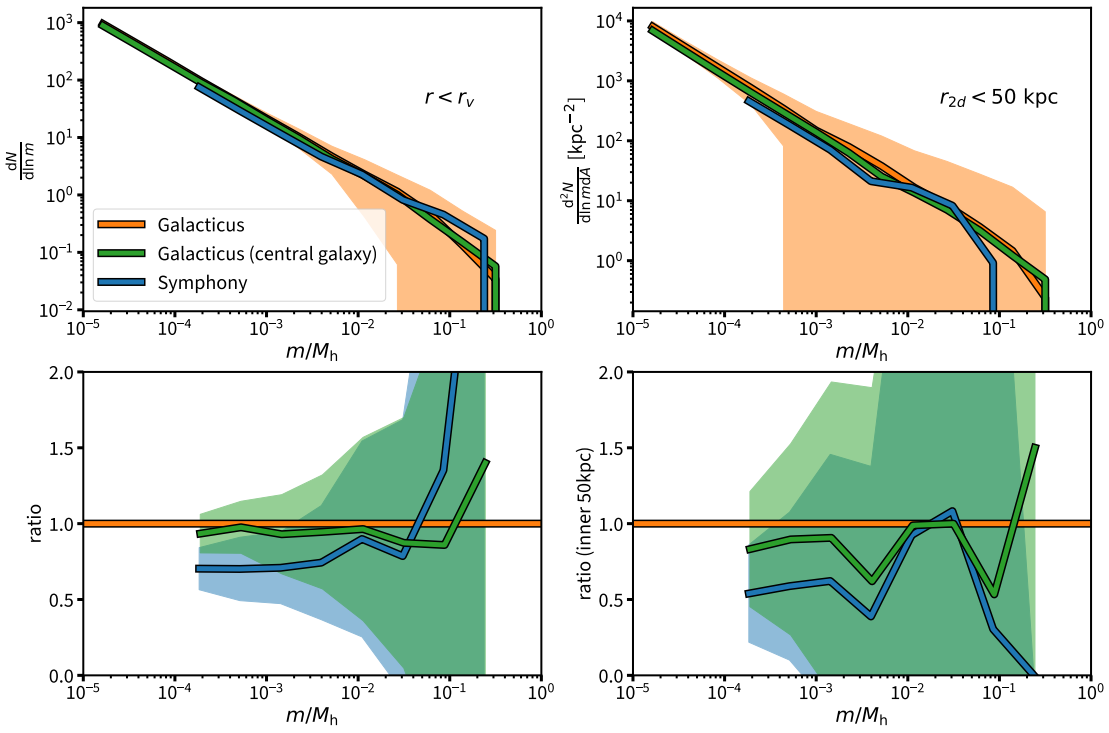


Figure 2.5: Upper left: the SHMF over the entire virial volume, right: the SHMF for subhalos projected within 50 kpc. Results are shown for group mass halos ($M_h \sim 10^{13}$ M$_\odot$) at redshift $z = 0.5$. The `galacticus` DMO predictions are shown in orange, `galacticus` predictions for the central galaxy are shown in green, and results for the Symphony suite are shown in blue. Lower left and right show the ratios to the `galacticus` DMO predictions. The effect of the central galaxy can be seen as a reduction in normalization of the SHMF no significant change to the logarithmic slope, at least for low mass halos. The `galacticus` and Symphony mass function and its halo to halo scatter are calculated by taking the mean and standard deviation of the value of the mass functions of every halo at every mass bin (in the upper panel, only the the `galacticus` DMO scatter is shown for clarity). For the lower panels, the values of the ratio and its scatter are calculated by dividing the average mass function and its scatter by the average value of the `galacticus` DMO mass function at each mass bin. Note that the difference in normalization for `galacticus` with and without a central galaxy is significantly less than the difference between our `galacticus` models and the Symphony dark matter only simulation suite.

Parameters	Description	Value	Halo-to-Halo Scatter	Units
$\Sigma_{\rm sub}$	Unevolved PSHMF amplitude	420	73.5	10^{-3} kp
$f_{\rm s}$	PSHMF tidal stripping coefficient	0.0282	0.0105	-
$f_{\rm m}$	Average subhalo mass loss fraction	0.0252	0.0095	-
$f_{\rm s} \cdot \Sigma_{\rm sub}$	Evolved PSHMF amplitude	11.6	3.99	10^{-3} kp
α	Unevolved logarithmic PSHMF slope	-1.94	-	-
$\alpha_{\rm b}$	Evolved logarithmic PSHMF slope	-1.97	-	-

Table 2.2: Table of best fits for the projected subhalo mass function (PSHMF, Eq. 2.7 and 2.8). Fitting coefficients are averaged over 224 host halos per redshift and host halo mass (22,400 host halos in total). Projections are taken within a $10-20$ kpc annulus, and averaged over 3 projections in the xy, yz, and xz planes. Due to the low number of halos in the aperture, halo to halo scatter for the SHMF slope cannot be calculated.

Simulation	M_h [$M_\odot$]	z	PSHMF Amplitude [$10^{-3}\times$ kpc^{-2}]	scaled [10
`galacticus`	$1.0 \cdot 10^{13.0}$	0.5	11	
PonosV	$1.2 \cdot 10^{13.0}$	0.7	6	
PonosQ	$6.5 \cdot 10^{12.0}$	0.7	6	
Milky Way Satellites Prior	$1.7 \cdot 10^{13.0}$	0.0	$1-2$	4
Symphony	$1.0 \cdot 10^{13.0}$	0.5	5	

Table 2.3: Amplitude of the evolved PSHMF, given in terms of the $f_{\rm s} \cdot \Sigma_{\rm sub}$ (see Eq. 2.8). In the "scaled" column, the number density has been extrapolated to a $10^{13} M_\odot$ halo at redshift $z = 0.5$ using the scaling relations presented in this work. A logarithmic slope of the subhalo mass function $\alpha = -1.97$ was assumed.

Model	k_1 (mass scaling)	k_2 (redshift scaling)
`galacticus` (unevolved)	0.55	0.37
host density profile	0.58	0.30
`galacticus` (evolved)	0.37	1.05
`galacticus` with central galaxy (evolved)	0.43	1.06
Han et al. [2016] ($\gamma = 0.8$)	0.37	0.77
Han et al. [2016] ($\gamma = 1.0$)	0.34	0.86
Han et al. [2016] ($\gamma = 2.0$)	0.30	1.12
Han et al. [2016] (γ Interp)	0.50	0.83

Table 2.4: The best fits to power law scaling relations predicted by our `galacticus` models and estimates using the Han et al. [2016] model. The upper section gives the scaling for the unevolved case, while the lower section gives the scaling results for the evolved case. We compare the scaling relation for the unevolved case for the scaling relation which would be obtained if the scaling was entirely determined by the host's density profile evolution (Eq. 2.1). For the evolved case, we compare the PSHMF scaling prediction to that of the Han et al. [2016] model (Eq. 2.5). The transfer function from the host's smooth dark matter density profile is parameterized as a power law, with logarithmic slope γ (Eq. 2.4). We consider two possibilities for γ, one where γ is a constant and the other with γ a function of halo mass. For the case of mass independent γ, we compute the scaling coefficients for a range of physically meaningful values of γ. Finally, to study the effect of γ varying systematically with mass, we linearly interpolate the best-fit γ values as a function of mass (Table 2.1) to compute the scaling relations ("γ interp").

2.4 Discussion

2.4.1 Subhalo Populations of a Halo at Fixed Mass and Redshift

Here we discuss properties of the subhalo populations of group mass ($M_h \sim 10^{13}$ $M_\odot$) halos at redshift $z \sim 0.5$. The `galacticus` predictions for the unevolved SHMF agree with findings of Van den Bosch et al. [2005], that the unevolved SHMF can be written in a universal form (Eq. 2.11). Furthermore, we find that `galacticus` agrees with the empirical formula of Gao et al. [2004], that for the unevolved case the SHMF scales as a function of host halo mass and the abundance of halos in the universe as a whole (Eq. 2.12). Comparing the `galacticus` and Symphony group halo suite predictions for the normalization of the subhalo mass function within the inner 50 kpc, we find that `galacticus` predicts on average nearly twice the subhalos in this region. Over the entire virial volume we find similar results, with Symphony predicting around 40% the number of subhalos as `galacticus`. We note that Symphony and `galacticus` agree within halo to halo scatter, and N-body simulations such as Symphony and Caterpillar disagree on the 25% level [Nadler et al., 2023]. We also find that the suppression of the subhalo mass function due to the potential of the central galaxy is negligible when compared to the current theoretical uncertainty. Similarly, we find that Han et al. [2016] [Han et al., 2016] is in reasonable, although not perfect, agreement with `galacticus` on the spatial distribution as well as the halo mass and redshift scaling in projection. Here, we provide further discussion for each of these results.

Here, we discuss the expectations for the spatial distributions from the Han et al. [2016] model and how they compare to predictions from `galacticus`. The Han et al. [2016] model describes the unevolved spatial distribution of subhalos as following the dark matter density distribution of the host halo. As shown in Figure 2.2, our `galacticus` models show a difference between the unevolved subhalo spatial distribution and the host's smooth dark matter density profile (normalized to 1 at $r = r_{\rm v}$).

At first glance, this is difficult to explain with dynamical friction, as the dy-

namical friction time scale is proportional to the subhalo to host halo mass mass ratio ($m/M_{\rm h}$) and is longer than age of the universe for sub-halos with with less than 1/20 the mass of the host halo [Boylan-Kolchin et al., 2008]. However, older subhalo will have fallen into a much less massive host (in which $m/M_{\rm h}$ is much closer to 1), and therefore they would experience a much smaller dynamical friction timescale.

To determine if dynamical friction is the cause of this difference, `galacticus` could be run with and without dynamical friction enabled. We leave this to future work. The Han et al. [2016] model does not include effects of dynamical friction, while the `galacticus` models does. For the evolved case, Han et al. [2016] use a power law profile to model the ratio of evolved spatial distribution to the host's dark matter density profile. Figure 2.2 shows that `galacticus` does not follow this expectation exactly, with deviations from a power law near the virial radius.

When comparing our `galacticus` models to the Symphony Group suite at redshift $z = 0.5$, we find that `galacticus` predicts a normalization of the projected SHMF a factor of ~ 2 times higher than Symphony. We note that this difference remains within 1σ halo to halo scatter and caution should be taken when interpreting this result due to the small number of subhalos in this volume which makes obtaining good statistics challenging. If the difference is statistically significant, there are several possible explanations for the discrepancy. A large body of works has drawn attention to artificial disruption present in N-body simulations [van den Bosch et al., 2017, Benson and Du, 2022, Green and van den Bosch, 2019, Errani and Peñarrubia, 2019]. An estimate for spurious suppression due to artificial disruption is 10–20% over the entire virial volume, only increasing to a maximum factor of 3 within the inner 2% of the virial radius [Benson and Du, 2022]. Another possibile explanation is inaccuracies in halo finding algorithms. In our work, we use results based on the Symphony subhalo catalogs extracted using the Rockstar [Behroozi et al., 2012] halo finder. Recent work using a new halo finding algorithm by Mansfield et al. [2024] has identified 15–40% more subhalos within the virial radius when compared to the Rockstar results, increasing to 35–120% within $r < r_{\rm v}/4$. Another possibility may be the `galacticus` physics models. Our

`galacticus` model does not include a model of pre-infall tidal stripping [Behroozi et al., 2014]. We do not make a determination if the discrepancy is numerical in nature and leave studying the underlying cause(s) of the differences to future work. We find subhalos of a group scale host that appear near the Einstein radius in projection on average retain $\sim 4\%$ of their infall mass after tidal stripping.

2.4.2 Subhalo Population Dependence on Host Properties

We discuss the scaling the projected SHMF as a function of halo mass and redshift. For the evolved case, our scaling relations from `galacticus` and our analytic models predict that for a factor of 10 increase in host halo mass the evolved PSHMF function will increase by a factor ~ 2. This is similar to the factor of ~ 3 used in the extrapolation performed in Xu et al. [2015], but differs from the factor of ≈ 8 used Gilman et al. [2019a]. Differences from Gilman et al. [2019a] are likely due to a change in `galacticus`, in particular an update to the treatment of higher order substructure. The change in higher order substructure treatment had a large impact on the spatial distributions, with the previous spatial distributions of subhalos being more cuspy.

Next, We compare our scaling results from `galacticus` to the Han et al. [2016] model of the spatial transfer functions. Three forms of the tidal stripping transfer function, $\hat{T}$ are considered (see Equations 2.5 and 2.12), with fits to the projected scaling relation fits provided in table 2.4. All cases considered result in host halo mass scaling similar to `galacticus`, with a 10 times increase in mass resulting in a roughly ≈ 2 times increase in the normalization of the PSHMF. Fixing $\hat{T}(r/r_{\rm v})$ (see equations 2.5 and 2.4) to be invariant in halo mass ($\gamma = 0.6, 1.0, 2.0$ and "`galacticus` interp" cases in Table 2.4) results in a best fit of $k_1 \approx 0.3$. The redshift scaling results (k_2) range from 0.83 to 1.12, which is in agreement with the $k_2 = 1.05$ predicted by `galacticus` . Allowing γ to vary with halo mass (linearly interpolating between $\gamma = 1.06$ and $\gamma = 0.67$) does not increase agreement with the `galacticus` models.

2.4.3 Impact of Central Galaxy

Finally, we discuss the impacts of the central galaxy. We find that the central galaxy impacts the subhalo mass function normalization on the less than 5% level over the entire virial volume and the less than 15% level within a projected aperture near the Einstein radius. We note that studies on the effects of an analytic potential on the subhalos in Milky Way like systems have found a more significant reduction in the subhalo mass function that what we have found here [D'Onghia et al., 2010, Webb and Bovy, 2020]. Webb and Bovy [2020] simulate individual dark matter only subhalos in evolving external tidal fields with and without baryonic disk and bulge components finding that the inclusion of these baryonic components reduces the number of subhalos by 65%. We highlight two possible reasons why the impact of a group scale massive central elliptical may be less significant than that of a Milky Way–like central galaxy. First, the ratio of central galaxy mass to halo mass is, on average, nearly a factor of two lower at the group scale [Behroozi et al., 2019]. Secondly, the systems are physically distinct: the Milky Way contains contains a disk component which is not found in massive ellipticals.

2.4.4 Recommendations for Future $\Sigma_{\rm sub}$ Priors

Here, we discuss our recommendations for future priors on the number density of subhalos within the Einstein radius. To ensure matching of observable quantities, we use our `galacticus` models to recommend a prior on the amplitude of the evolved halo mass function $f_{\rm s}\cdot\Sigma_{\rm sub}$, instead of $\Sigma_{\rm sub}$ directly. Our `galacticus` models predict an average evolved number density of $f_{\rm s}\cdot\Sigma_{\rm sub}\sim 10^{-2}$ kpc^{-2}, while the Symphony suite predicts $f_{\rm s}\cdot\Sigma_{sub}\sim 5\cdot 10^{-3}$ kpc^{-2} (additional results are given in Table 2.3). Here, we do not investigate the possible reasons for the differences, and instead treat each result as equally likely so we can estimate a theoretical uncertainty. Across works compared here, we find that the value of $f_{\rm s}\cdot\Sigma_{\rm sub}$ varies from 10^{-3}–10^{-2} kpc^{-2}, spanning approximately an order of magnitude, with more recent works in the $5\cdot10^{-3}$–10^{-2} kpc^{-2} range. In a 100 kpc^2 aperture, this approximately corresponds to 0.1 to 1 subhalos with bound mass $8<\log_{10}(m/{\rm M}_\odot)\leq 9$ (or equivalently a number density of 10^{-3}–10^{-2} kpc^2). For reference, an average

Einstein radius of 1 arcsecond at redshift $z = 0.5$ corresponds to a ~ 150 kpc^2 aperture. We note that this is well within the allotted uncertainty of previous lensing studies. Furthermore, we estimate that the impact of a central galaxy (reduction in $f_{\rm s} \cdot \Sigma_{\rm sub}$ by a factor of $\sim 10\%$) is negligible compared to the current theoretical uncertainty between different models.

We recommend a prior centered around $f_s \cdot \Sigma_{\rm sub} = 10^{-2}$ kpc^{-2} and allowing for at least a factor of 2 (~ 0.3 dex) uncertainty. We note that this is a similar prior to the tighter prior considered by Gilman et al. [2024].

When Gilman et al. [2024] compared dark matter inferences using a 1.5 dex log-uniform prior (similar to that used in previous lensing studies) with an inference using a tighter Gaussian prior with width of 0.2 dex, the lower bound on the half-mode-mass improved by 2 dex. We recommend the continued use of a log-uniform prior as N-body simulations have consistently lower values of $f_s \cdot \Sigma_{sub}$ when compared to our `galacticus` models; a log-uniform prior treats both cases as equally likely. By implementing a 0.3 dex log-uniform prior we expect similar improvements on constraining power to those observed by Gilman et al. [2024].

2.5 Summary

In this work we study predictions for the population level statistics of subhalos of group mass halos at small projected distances. Particular focus is placed on the normalization of the projected SHMF, where we give recommendations for a more informed prior on this quantity than previous lensing studies. Here we provide a summary of key results in our paper:

- We present a new suite of `galacticus` models of the substructure of host halos with masses $12 \leq \log_{10}(M_h/{\rm M}_\odot) \leq 13.5$ and redshifts $0.2 \leq z \leq 0.8$.
- Using these simulations, we measure the projected number density of subhalos near the Einstein radius. The scaling in halo mass and redshift is well described with a power law with coefficients $k_1 = 0.55$ (halo mass) and $k_2 = 0.37$ (redshift) for the unevolved case and $k_1 = 0.37$ and $k_2 = 1.05$ for the evolved case.

The projected scaling differs from the scaling of the total number of satellites within the virial volume (total subhalo count scales nearly linearly with the mass of the host and is almost independent of redshift in both the evolved and unevolved cases).

- The evolved and unevolved projected SHMF amplitude scales differently as a function of halo mass and redshift due to different spatial distributions between the two cases. For the evolved case, the projected SHMF amplitude approximately scales with the cube root of the host halos mass and nearly linearly in redshift. We find the scaling of the PSHMF in redshift and mass to be in excellent agreement match between simulations and analytic models.

  Our analytic models show the scaling of the PSHMF can be explained in terms of two factors: 1) the scaling of the total number of subhalos within the virial volume of the host and 2) the spatial distribution of subhalos.

- On the group scale, a central galaxy reduces the normalization of the evolved SHMF by $< 5\%$ over the entire virial volume, increasing to 15% for the PSHMF within the inner 20 kpc. This is much less than the current theoretical uncertainty between different models/N-body simulations.

  For future lensing studies with group mass lenses, we find that the effects from a central galaxy have a negligible impact on the strength of tidal stripping experienced by subhalos and does not need to be considered when setting a prior $\Sigma_{\rm sub}$.

- We find subhalos of a group scale host that appear near the Einstein radius in projection on average retain $\sim 2\%$ of their infall mass after tidal stripping.

- We compare the projected number density of subhalos in a variety of N-body simulations to our `galacticus` models. We find that all models/simulations considered agree on the projected SHMF normalization well within a factor of ~ 2. Additionally, we find that `galacticus` and Symphony agree within the 1σ halo to halo scatter for subhalos halos in the bound mass range $9 < \log_{10}(m_b/\mathrm{M}_\odot) \leq 10$.

We recommend using a stronger prior on $f_s \cdot \Sigma_{\rm sub}$ when compared to previous lensing studies; specifically a log-uniform prior centered around $f_s \cdot \Sigma_{\rm sub} = 10^{-2}$ kpc^{-2} and allowing for at least a factor of 2 uncertainty. Compared to previous lensing studies, we expect a significant reduction in uncertainty on parameters like half-mode-mass when using this tighter prior.

Acknowledgments

We thank Ethan Nadler and Risa Wechsler for thoughtful comments and suggestions.

This research was conducted using Pinnacles (NSF MRI, # 2019144) at the Cyberinfrastructure and Research Technologies (CIRT) at University of California, Merced.

This research was supported in part by grant NSF PHY-2309135 to the Kavli Institute for Theoretical Physics (KITP).

AN and CG acknowledge support from the NSF through AST- 2206315 "Collaborative Research: Measuring the physical properties of DM with strong gravitational lensing" and through JWST-GO program #2046 which was provided by NASA through a grant from the Space Telescope Science Institute, which is operated by the Association of Universities for Research in Astronomy, Inc., under NASA contract NAS 5-03127.

DG acknowledges support for this work provided by the Brinson Foundation through a Brinson Prize Fellowship grant, and from the Schmidt Futures organization through a Schmidt AI in Science Fellowship.

2.A Evolution of the Central Galaxy

In our models with a central galaxy, we model the evolution of the potential of a central massive elliptical using empirical relations from Behroozi et al. [2019]. We plot properties of the central galaxy in our `galacticus` models in Figure 2.6.

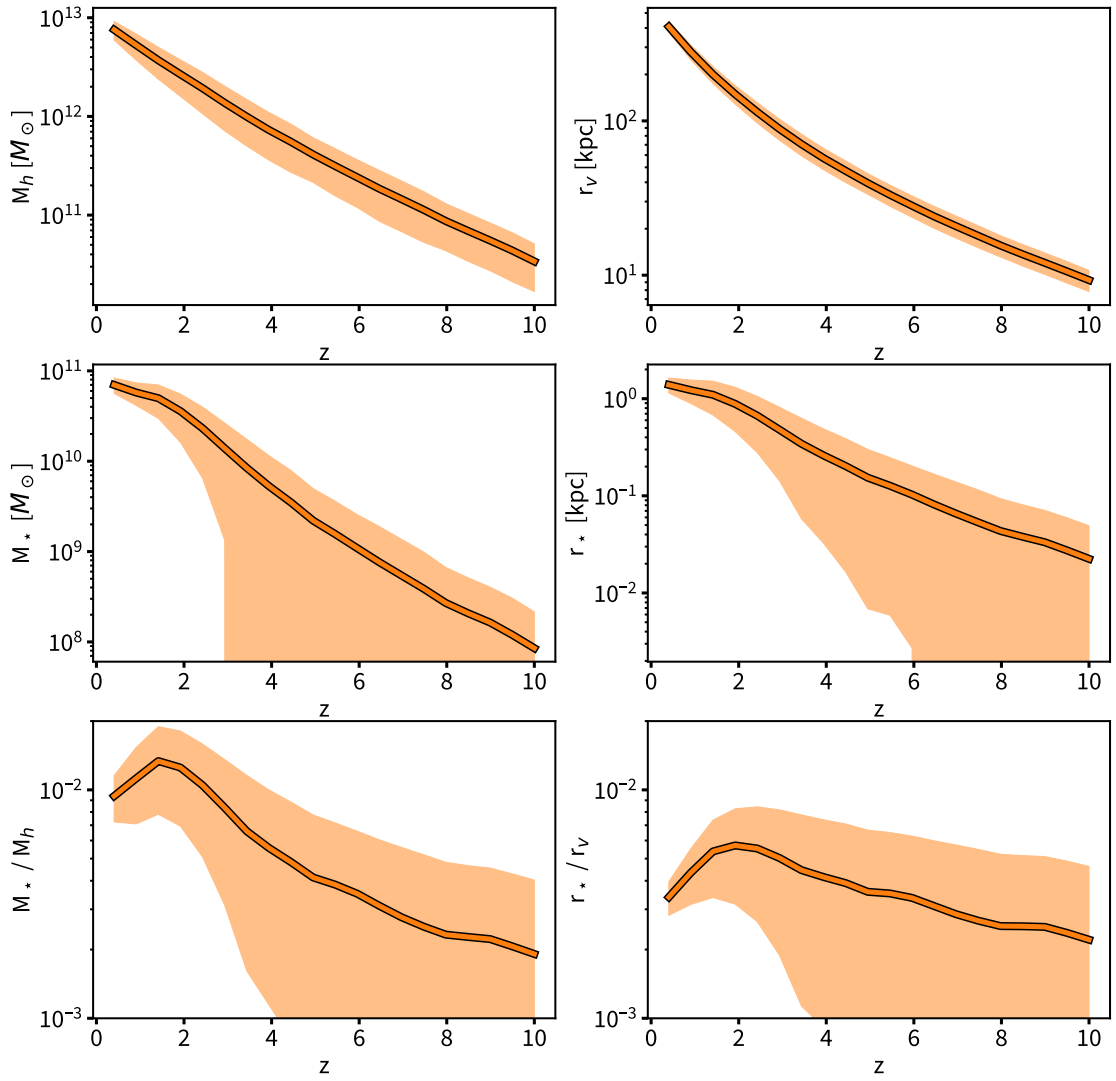


Figure 2.6: Our `galacticus` model predictions for the evolution of a $10^{13}\mathrm{M}_\odot$ halo and its central galaxy. Stellar masses are determined using the Behroozi et al. [2019]. relation between halo mass and stellar mass, while galactic radii are determined using empirical fits to the stellar mass to stellar radius relation provided by Shen et al. [2003]. For simplicity, no scatter has been included in either of these relationships. The top left panel gives the evolution of the mass of the central halo while the top right panel gives the evolution of the virial radius of the halo. The middle left panel shows the evolution of the stellar mass, while the middle right panel shows the evolution of the Hernquist radius of the central galaxy. Finally, the lower left and right panels show the evolution of the ratio of stellar mass to halo mass and Hernquist radius to virial radius, respectively. The curve shows the mean value, while the shaded region shows 1σ scatter.

Chapter 3

T1DES in SIDM: A New Class of SIDM Subhalo Simulations

This chapter is an initial draft of a work that we anticipate will be submitted for publication. We thank our collaborators Andrew Benson, Xiaolong Du, Daniel Gilman, Ethan Nadler, Andrew Robertson and Ryan Keeyley for their helpful input throughout the making of this manuscript. This manuscript has not has not yet been sent to our collaborators for review and aproval.

Abstract

Dark matter models with self interactions have gained to resolve apparent shortcomings of the LCDM cosmology. Testing these models of Self Interacting Dark Matter (SIDM) requires accurate modeling the gravothermal evolution of halos under accelerated heat transport facilitated by these interactions. The picture becomes more complicated for subhalos (halos evolving within the tidal field of the host), as they have their gravothermal evolution altered by tidal physics in addition to any potential dark matter self interaction; many probes of SIDM such as gravitational lensing sample from a halo population that includes subhalos. Existing methods of simulating SIDM subhalos are very computationally expensive or make restrictive assumptions on tidal physics. To address these shortcomings, we present `T1DES` (Tidal 1D Evolution of SIDM subhalos), a new class of 1D SIDM subhalo

simulations incorporating realistic tidal physics while maintaining the many orders of magnitude speed up over full 3D SIDM simulations. We find that under no evaporation the core collapse is accelerated, with nearly 4 times as many subhalos collapsing compared to isolated counterparts.

3.1 Introduction

Cold Dark Matter or CDM has had a long string of successes explaining the largest scales of cosmology, from the cosmic microwave background Planck Collaboration [2020b] to the spatial and mass function distributions of galaxies White and Rees [1978], White and Frenk [1991], De Blok et al. [2008], Weinberg et al. [2015]. However, recent inferences on the sub-galactic from direct gravitational imaging the potential of lenses such as the jackpot lens (SDSSJ0946+1006) Vegetti et al. [2010], Minor et al. [2021], B1938+666 and J0946+1006 Tajalli et al. [2025] have provided evidence for anomalously dense (high concentration) dark matter halos compared to CDM predictions. Although these results have been hotly debated and solutions within the CDM framework have been proposed He et al. [2025], the intriguing possibility remains that these results hint at physics beyond CDM. A class of alternate dark matter models with non-negligible self-interaction on the sub-galactic scales, called self-interacting dark matter (SIDM) has been proposed as an explanation of anomalously dense halos inferred in these systems Zhang et al. [2025]. In these models, dark matter self-interaction facilitates the conduction of heat within the halo; due to the negative heat capacity of self-gravitating systems, runaway heat transfer can occur leading to the halo gravothermally collapsing, leading to extremely dense collapsed remnants at the end of their evolution (see Refs. Tulin and Yu [2018], Adhikari et al. [2025] for reviews).

In this paper we focus on studying core collapse rates of SIDM subhalos; lensing and stellar streams results sample from a population that includes subhalos. To infer SIDM constraints, it is important to understand how fast gravothermal collapse may occur given an SIDM model; tidal forces from the host are know to alter core collapse rates Zeng et al. [2022], Shah and Adhikari [2024], Nishikawa

et al. [2020], Nadler et al. [2025b], Yang et al. [2024]. Previously there have been 2 main methods to study SIDM subhalos: 3-D N-body simulations (both cosmological Vogelsberger et al. [2012], Rocha et al. [2013], Dooley et al. [2016], Nadler et al. [2020], Turner et al. [2021], Ebisu et al. [2022], Yang et al. [2023a], Nadler et al. [2025a] and idealized Kahlhoefer et al. [2019], Zeng et al. [2022, 2023], Zhang et al. [2025], Klemmer et al. [2026]) and gravothermal fluid modeling Nishikawa et al. [2020], Shah and Adhikari [2024]; while these have all been fruitful ways of studying SIDM substructure in their own right, they each have significant drawbacks. 3-D N-body simulations, which are traditionally the primary method of studying dark matter halos are computationally expensive especially in the core collapse regime; tidal mass loss amplifies this challenge by requiring a larger number of particles to resolve the halo initially, often requiring tens of thousands of core hours to properly resolve core collapse even in idealized simulations. Gravothermal fluid models, while much computationally cheaper than 3-D N-body simulations use restrictive assumptions on tidal physics namely approximating tidal stripping as one or more truncations event where mass is abruptly removed from the halo. Because of these limitations, accurately predicting the core collapse rates for large samples of halos across different SIDM models remains a challenge.

In this work, we develop `T1DES` (Tidal 1D Evolution of Subhalos): a pipeline to simulate the evolution of idealized SIDM subhalos in 1-D with realistic tidal evolution. By simulating subhalos evolution in 1-D, we are able to simulate SIDM physics much more computationally efficiently than the 3D case, resulting in over an order of magnitude speed up in some case enabling previously infeasible exploration of SIDM model space. We begin by explaining our methods in Section 3.2. First, we discuss generating a realization of `galacticus` Benson [2012] CDM halos in Section 3.2.1 to set the initial conditions for our 1-D simulations and provide a target mass loss histories. Next, in Section 3.2.2 we discuss how we modify the dynamics of the 1-D simulation code `NSphere` Kamionkowski et al. [2026], Kamionkowski and Sigurdson [2025, 2026] to include an effective tidal field calculated from a target mass loss history. Then in Section 3.2.3 we discuss the SIDM physics implemented in `NSphere` along with the several modifications we

make for our use case including a dynamic timestep and core collapse termination condition. We then discuss the results of our simulations in Section 3.3 and provide a summary and concluding remarks in Section 3.4.

3.2 Methods

We aim to explore the population level statistics expected of subhalos systems hosted by strong gravitational lenses. To accomplish this we develop the `T1DES` pipeline to simulate realizations of SIDM subhalos across various tidal fields and under the influence of a tidal field. We use a modified version of the 1-D N-body `NSphere` code, this provides a large speed up over traditional 3-D N-body simulations; the slowest halos in our pipeline complete in ~ 600 core hours while a comparable idealized 3-D SIDM simulation can take $> 10,000$ core hours to complete. In this section, we give an overview of the `T1DES` pipeline with a visual diagram given in Figure 3.1. Our pipeline requires the the mass history, infall properties (i.e scale radius r_s, virial radius r_v, and time since infall t_{inf}) and tidal tracks (i.e $R_{\texttt{max}}(t), V_{\texttt{max}}(t)$) [1]. In Section 3.2.1 we discuss generating generating realizations of lens like hosts and their subhalos in `galacticus` to obtain these quantities. After generating the realizations in `galacticus` we proceed to evolve the halos in CDM mode in our modified version of `NSphere` . While 1-D N-body simulations provide a significant speed up of full 3-D N-body simulations they provide a unique challenge: simulating asymmetric dynamics of a tidal force in 1-D. Instead of calculating a tidal field directly from the subhalo's orbit, we do a dynamic "first pass" to calculate the force at every timestep required to match a subhalo's mass history; then we rerun the subhalo with a smoothed version of this tidal field. We discuss this as well as the dynamics implemented in our simulations in Section 3.2.2. Finally, we run `NSphere` in SIDM mode using a smoothed tidal field; we give an overview of the SIDM implementation in Section 3.2.3.

[1] Tidal tracks are only required for SIDM mode, as they are used in setting the simulation termination conditions

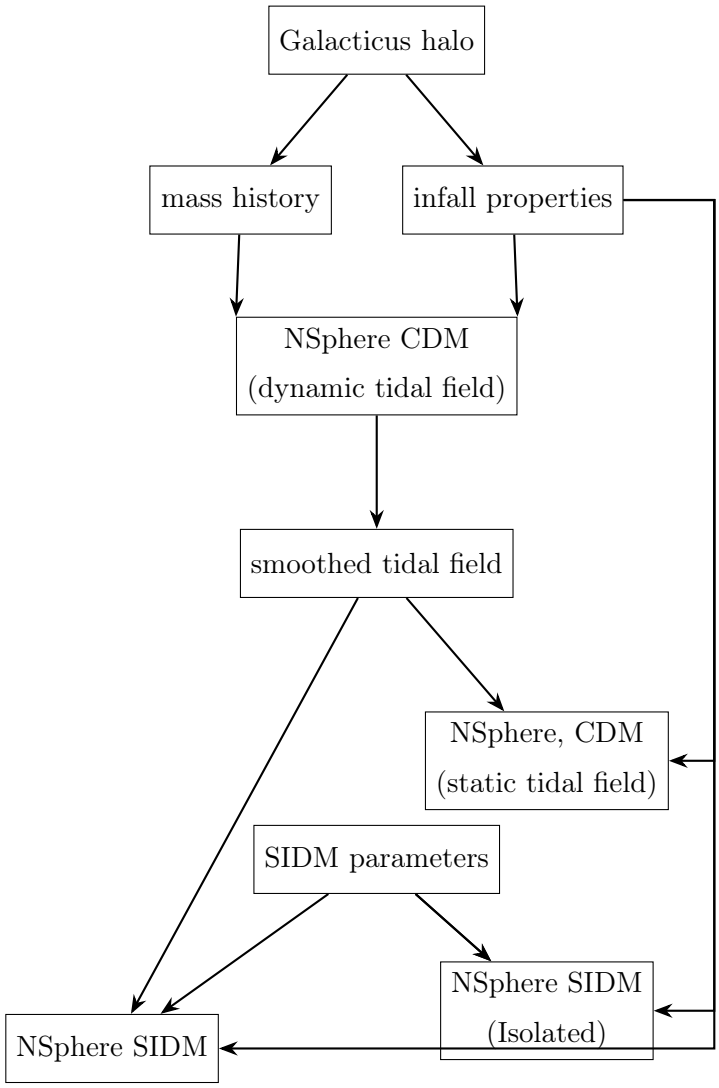


Figure 3.1: A diagram of the `T1DES` pipeline simulating a single subhalo. The `T1DES` pipeline begins with extracting infall properties (r_s, r_v, z_{infall}) as well as the bound mass of the subhalo as a function of time. Our main results use the `galacticus` galaxy formation model to extract the infall mass properties of the subhalo, however we also include results using properties extracted from N-body simulations. These are used to set the initial conditions of the halo and begin running a first pass of NSphere in "dynamic" tidal field mode. During this run, NSphere calculates a tidal field "in flight" that results in a mass history approximating that of the galacticus halo. After the run, the tidal field is then smoothed on the scale of the dynamical time of the subhalo to remove the rapid variations in the initial calculation of the tidal field. For comparison, we rerun NSphere in CDM mode to get a "replay" of the subhalo's evolution using the smoothed tidal field. Then we run NSphere in SIDM mode with the smoothed tidal field. Finally, for comparison we run a NSphere SIDM on an isolated halo corresponding to the infall properties of the subhalo.

3.2.1 Subhalo Realizations

We aim to simulate realizations of lensed subhalos in a $10^{13} M_{\odot}$ host in SIDM; our T1DES pipeline requires subhalo properties (mass history, infall properties) as a starting point. To obtain these quantities we simulate subhalo populations we use the semi-analyic galaxy formation model, `galacticus` to generate subhalo populations. Semi-analytic are faster than traditional 3D N-body simulations and allow us to realize subhalo populations quickly. The `galacticus` model is particularly well suited for our use case because of it's recent calibration to high resolution idealized simulations [Du et al., 2024] and it's extensive and well tested library of dark matter physics. However, we note that our model is not limited to only using subhalo properties obtained using the `galacticus` galaxy formation model. It is possible to use evolutionary histories obtained from other sources such as N-body simulations, which we explore later in this paper.

We give a brief summary of the `galacticus` dark matter models used and then discuss the simulation configurations used and selection criteria for subhalos. First we discuss the discuss the procedure `galacticus` uses to simulate subhalo populations: first, merger trees are generated and then evolved forward in time using a set of analytic and semi-analytic models. Finally, we discuss our simulation parameters and the selection criteria for lensing like subhalos.

Merger tree generation — We build our `galacticus` merger trees algorithms presented in Cole et al. [2000b] and Parkinson et al. [2008]. The algorithm is works by calculating branching probabilities: step back in time and at each time step calculating branching probabilities using the extended Press and Schechter [1974] formalism. Using the branching probabilities, a merger tree is then built backwards in time that includes both merger events (infalling halos) and smooth accretion. Halo scale radiis are set using the Ludlow et al. [2016] and concentrations are set using the Diemer and Joyce [2019] model.

Node evolution — After merger trees are built, our `galacticus` model tracks infalling (subhalos) halos and forward evolves them in time; first subhalo's orbits are initialized, then their orbits and profiles are evolved using a set of analytic models. When a subhalo infalls (a merger event happens) it's orbit is initialized

according to the potential of the host. After initialization the subhalo's orbit is evolved forward in time using models for dynamical friction, tidal heating and tidal stripping. We give a high level overview of these models here:

- Dynamical friction — Dynamical friction causes subhalo orbits to decay; our configuration uses the Chandrasekhar [1943] model with a Columb logaritm $\log \lambda =?$ tuned to the Griffen et al. [2016] simulations.

- Tidal Heating — Interactions with the tidal field of the host dynamically heat the host's dark matter density causing the profile to expand. Our `galacticus` models use the Gnedin et al. [1999] tidal heating model tuned to high resolution idealized simulations provided by Du et al. [2024].

- Tidal Stripping — Mass outside of the tidal radius is removed by tidal forces. Our `galacticus` models use the Zentner et al. [2005] tidal stripping model with and is tuned to high resolution idealized simulations provided by Du et al. [2024].

Simulation Configuration — Using the `galacticus` model we simulate the subhalo population of 10 lensing like $10^{13} M_\odot$ halos. In this work, we focus on subhalos in the mass range $10^9 \leq m/M_\odot \leq 10^{10}$. Our `galacticus` models have an infall mass resolution of $8.00 \cdot 10^8$ and subhalos are destroyed is they lose more than 90% times their infall mass due to tidal effects. For our T1DES simulations, we select "lensing like" subhalos: subhalos with a $20 < r_{\texttt{2d}}/\texttt{kpc} < 50$ projection of the center of the host. These are subhalos that are likely to be sampled from in a strong lensing study. We also exclude subhalos that have been in the subhalo for less than 1.0`Gyr` as they likely have not completed a single orbit. Subhalos within x kpc of the host are destroyed.

For this analysis we only first-order substructure; that is subhalos that are field halos until they fall into the primary host. This contrasts with 2nd order and higher subhalos which are accreted into intermediate halos before infalling into the host.

- Extreme mass histories — `galacticus` often predicts extreme early mass loss histories for 2nd order and higher subhalos. This is likely an artifact of the

way `galacticus` handles the orbits of these subhalos; these extreme mass histories characterized by rapid early mass loss when accreted into the first host and much steadier mass loss when accreted into the final host. This is likely an artifact due to how our `galacticus` handles subhalo orbits: with 2nd order substructure disabled some subhalos rapidly loosing mass on first accretion then when the first host is acrreted, the subhalo abruptly stops loosing mass to it's first host and begins loosing mass to the it's new host. Our pipeline often disagrees with `galacticus` on the internal properties of these subhalos. We leave further investigation to future work.

- Simplifying our analysis — In our `galacticus` models, 2nd order subhalos have 2 distinct orbits: one in the precursor subhalo and one in the primary host; this would significantly complicate our analysis. For the purpose of this work, we study the simpler case of 1st order substructure and leave these complexities to future work.

Due to these complications, we leave exploring extending the `T1DES` pipeline to higher order substructure to future work.

3.2.2 Gravitational Dynamics

We modify the 1D N-body code `NSphere` to include the influence of a linear tidal force. The tidal dynamics experienced by a subhalo's particle are asymmetric in our subhalo's coordinate frame. To model tidal physics in our spherically symmetric simulations, we need to choose an effective tidal field; we choose our tidal field to match the target halo's bound mass loss rate. Choosing a tidal field strength can be thought of as choosing a tidal radius. We would expect mass outside the Roche volume of the subhalo to be lost on order the dynamical time of the subhalo. Our choice to match the mass loss history of the subhalo is natural: where the effective tidal force dominates the gravitational subhalo, the subhalo loses mass; this forms a volume () outside of which mass is being lost; we choose a tidal radius enclosing approximately the same mass as this volume.

Under the influence of gravity and a linear tidal force with the assumption of

spherical symmetry, the acceleration of ith shell is

$$\ddot{r}_i = -\frac{GM(< r_i)}{r_i^2} + \frac{l_i^2}{r_i^3} + f_t r_i \tag{3.1}$$

, where r_i is the distance from to center of the subhalo from the ith particle, l_i is the angular momentum per unit mass on each particle, $M(< r_i)$ is the mass enclosed by radius r_i and f_t is the linear tidal field. For each `T1DES` halo, we calculate a aim to calculate a tidal field to approximate the mass loss history of it's corresponding galacticus subhalo. Following Du et al. [2024], outside the tidal radius, we assume the rate of mass loss to be on the dynamical timescale of the halo

$$\frac{dM_{\rm sub}}{dt} = -\alpha_s \frac{M_{\rm sub}(> r_t)}{T_{\rm dyn}} \tag{3.2}$$

where $M_{\rm sub}$ is the gravitationally bound mass of the subhalo, α_s controls the tidal stripping rate, r_t is the tidal radius, and $T_{\rm dyn} = 2\pi\sqrt{r_t^3/16GM_{\rm sub}(< r_t)}$ is the dynamical time of the subhalo. For a `T1DES` subhalo with bound mass M_T at time t we solve the following equation:

$$-\alpha_s \frac{M_{\rm T}(t, > r_t)}{T_{\rm dyn}} = \min\left\{\frac{M_{\rm g}(t+\Delta t) - M_{\rm T}(t)}{\Delta t}, 0\right\} \tag{3.3}$$

to calculate a tidal radius that results the reduction in mass to the mass galacticus halo, M_g at time $t + \Delta t$; we do not allow the right hand side of the equation to be greater than zero since the subhalo can only lose mass. Due to the self-correcting feedback introduced by the right hand side of Eq. 3.2, we find our simulations are fairly insensitive to our choice of α_s, we choose $\alpha_s = 4$ to match Du et al. [2024]. The tidal force and gravitational force balance at the tidal radius, r_t; assuming a linear tidal force $F_t \propto r$ the tidal field corresponding to a tidal radius r_t is

$$f_t = \frac{GM(< r_t)}{r_t^3} \tag{3.4}$$

. Since our prescription can produce large jumps in the tidal field, we smooth the tidal field on order the the dynamical time of the subhalo, T_{dyn} in post processing; we find that running with the smoothed tidal field using a gaussian filter with width $1/10T_{dyn}$ produces nearly identical halo evolution to the original run.

We show the evolutionary history for both stages of our `T1DES` model as well as it's `galacticus` counterpart in Figure 3.2.2. Due to our choice of tidal field,

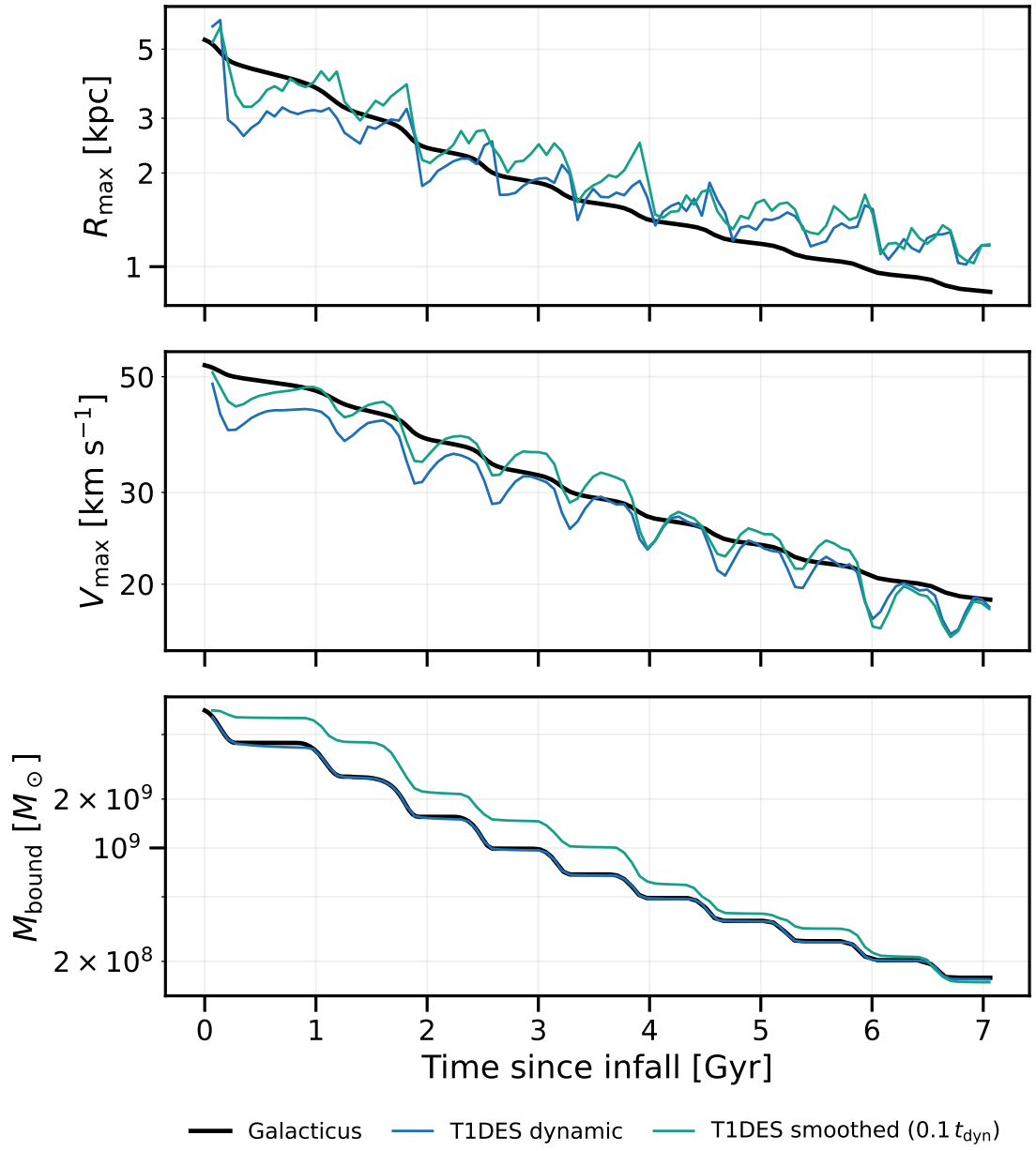


Figure 3.2: The evolutionary history of an example CDM `T1DES` subhalo and it's `galacticus` counterpart. We show the bound mass history, $M_{\texttt{bound}}$ as well as the internal properties of the subhalo, $R_{\texttt{max}}$ and $V_{\texttt{max}}$ (see Eq. 3.7). Each sudden drop in the mass history corresponds to a peri-centric passage of the subhalo, this subhalo completes 10 pericentric passages in it's 7Gyr evolution within the host, losing $\sim 90\%$ of it's mass since infall. Results from both `T1DES` stages are shown: results from the `T1DES` dynamic stage, where the is dynamically calculated as well as the results from the `T1DES` dynamic stage, where we rerun with the dynamically calculated tidal field smoothed by a gaussian kernel. We note that while our model calculates a tidal field to match the provided `galacticus` mass history, it is not tuned to match the internal $R_{\texttt{max}}$ and $V_{\texttt{max}}$ properties; even so it is in good agreement with `galacticus` for nearly all halos.

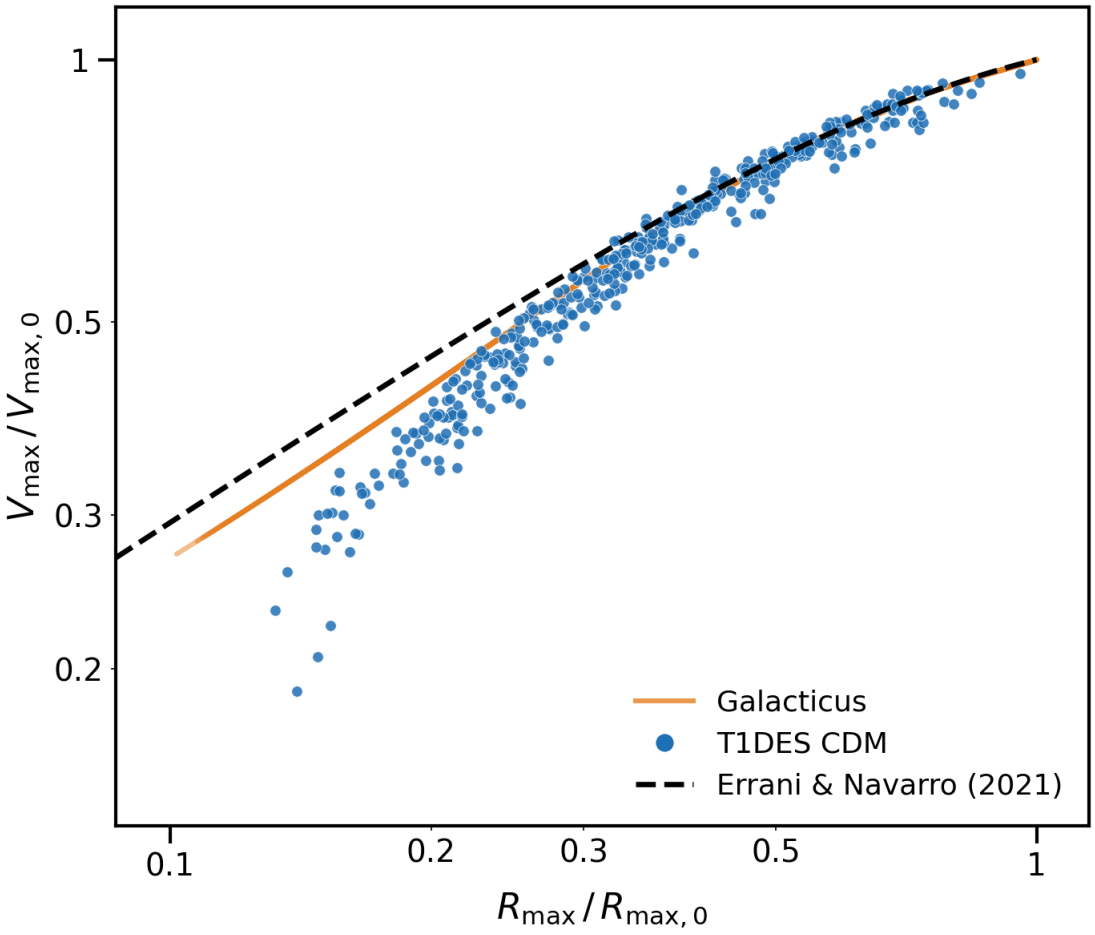


Figure 3.3: We plot the normalized tidal tracks from our `T1DES` CDM smoothed stage simulations, our `galacticus` models and from the empirical Errani and Navarro [2021]. For each subhalo in our suite, we plot the $R_{\mathtt{max}}/R_{\mathtt{max,0}}$ and $V_{\mathtt{max}}/V_{\mathtt{max,0}}$ at the subhalo's apocenter where $R_{\mathtt{max,0}}$ and $V_{\mathtt{max,0}}$ are the initial $R_{\mathtt{max}}$ and $V_{\mathtt{max}}$ of the subhalo at apocenter. We plot the Normalized tidal tracks from `galacticus` , `T1DES` and predictions . When properly normalized, CDM tidal tracks are expected to follow a near universal relation. We find general good agreement on the tidal track level between our simulations; with slight disagreement for low $R_{\mathtt{max}}$ and $V_{\mathtt{max}}$.

in CDM, our model is in excellent agreement with the bound mass history of both `galacticus` and 3D idealized N-body simulations. In addition, we also compare our model's prediction for the internal properties $R_{\texttt{max}}$ and $V_{\texttt{max}}$ curves to `galacticus` and predictions from idealized N-body simulations with the caveat that our model sometimes overshoots the mass loss from `galacticus` simulations. Figure 3.2.2 shows an important diagnostic for our simulations: we plot the normalized tidal tracks, $R_{\texttt{max}}/R_{\texttt{max,0}}$ vs $V_{\texttt{max}}/V_{\texttt{max,0}}$ for each subhalo at apocenter. We include `T1DES` CDM model, along with results from out `galacticus` model and the empirical Errani and Navarro [2021] model; These curves are particularly important as they are the scale on which SIDM evolution happens Yang et al. [2024]. When properly normalized, CDM tidal tracks are expected to follow a near universal relation. We find general good agreement on the tidal track level between our simulations; with slight disagreement for low $R_{\texttt{max}}$ and $V_{\texttt{max}}$, however we note that this discrepancy is nearly as large as the difference between the `galacticus` model and Errani and Navarro [2021]. Additionally, the `galacticus` tidal tracks results form nearly a perfect curve with very little scatter; while our `T1DES` model has more significant scatter: likely due to the noise discrete noise inherent to N-body simulations. While our model is not specifically calibrated to match these curves, our model is in reasonable agreement with both the results from `galacticus` and 3D N-body simulations.

3.2.3 SIDM Physics

Here, we give an overview of the SIDM physics implemented in our simulations; we use the default NSphere SIDM implementation with the addition of an adaptive timestep. We discuss our choice of cross sections and finally, we discuss the termination conditions we impose on the simulation and how we define core collapse.

SIDM Scattering — The probability of the i th particle scattering in the simulation within some time scale Δt is evaluated by summing over all particles from

r_1 to $r_1 + \Delta r_i$

$$P_1(r_1, v_1, c_{\theta_1}) = \frac{\Delta t}{2}\left(\frac{m_p}{4\pi r_1^2 \Delta r}\right)\sum_{r_i=r_1}^{r_1+\Delta r}\sigma_{\text{eff}}(v_{\text{rel},i})v_{\text{rel},i}, \tag{3.5}$$

where m_p is the particle mass, v_{rel} is the relative velocity between the two particles and $\sigma_{\text{eff}}(v_{\text{rel}}) = \frac{1}{m}\int_{-1}^{1} d\cos\alpha\,(d\sigma/d\cos\alpha)$ is the total cross section with m being the mass of the elementary particle,. We use the default NSphere bin width, $\Delta r = r_{i_i+j} - r_{i_1}$ with $j_1 = 10$.

After particles scatter, their kinematics must be calculated. When two identical particles with velocities $\mathbf{v_1}$ and $\mathbf{v_2}$ scatter, the Center of Mass (COM) frame has velocity $(\mathbf{v_1} + \mathbf{v_2})/2$. COM frame, the particles have outgoing velocities $v_{\texttt{cm1}}$ and $v_{\texttt{cm2}}$ both having equal magnitudes to half the incoming relative velocity $v_{\texttt{rel,2}}/2$ and offset by angle α. We assume isotropic scattering; therefore, the cosine of the scattering angle, $\cos\alpha$ is chosen uniformly from -1 to 1 and the polar angle ϕ_f around $v_{\texttt{rel,2}}$ is chosen uniformly on the interval 0 to 2π. After the velocities have been calculated, they are projected into the halo frame and saved for the next timestep.

Timestep — The key timescale for SIDM evolution is the scattering time, $t_{\texttt{scatter}} = [\rho\sigma_{\texttt{m}}v]^{-1}$. To accurately model this evolution, the chosen timestep of the simulation, Δt_{step} must be much less that the scattering time ($\Delta t_{\texttt{step}} < t_{\texttt{scatter}}$) Therefore, the size of the timestep sets the maximum density (and velocity) that can be resolved by the simulation; by default, `NSphere` implements a constant timestep, $\Delta t_{\texttt{step}} = C$. In this method, the fixed timestep size is set by choosing a maximum resolvable density. However, this is inefficient as a very small timestep must be chosen to resolve the high density of core collapse and the halo will only reach this density at the very end of it's evolution. Instead, we implement an adaptive timestep; we choose the size of the adaptive timestep based on the probability of two particles scattering in the previous timestep, $P_{\texttt{i}}$ (see Eq. 3.5)

$$\Delta t_{\texttt{step,n+1}} = \max\left\{\epsilon\cdot\frac{\Delta t_{\texttt{step,n}}}{\max\{P_{\texttt{i}}\}}, \Delta t_{\texttt{step,min}}\right\} \tag{3.6}$$

where we ensure that if the particles maintained the same scattering probability in the next timestep, the maximum scattering probability would be ϵ. For our

simulations, we find good convergence with $\epsilon = 1.0 \cdot 10^{-4}$. Taking the minimum of all particle scattering probabilities can result in very small time steps, for instance due to a single pair of particles that are highly likely to scatter in the next timestep; we prevent these extremely small timesteps by implementing a timestep floor $\Delta t_{\texttt{step,min}}$. We set the timestep floor proportional to the maximum density we expect to resolve in the simulation, $\Delta t_{\texttt{step,min}} = [\bar{\rho}_{\texttt{max}}(< R_{\texttt{max}})\sigma_{\texttt{m}} \max\{V_{\texttt{max}}\}]$ where $\bar{\rho}_{\texttt{max}}(< R_{\texttt{max}})$ is the termination density of our simulations and $V_{\texttt{max}}$ is the circular velocity predicted for the target halo (in this case, we use the result from `galacticus` model). We conveniently use $\{V_{\texttt{max}}\}$ as the characteristic velocity because for the cross-sections studied and the termination condition we use, the termination $V_{\texttt{max}}$ in SIDM remains slightly higher than (maximum $\sim 20\%$ more) or less than the $\max\{V_{\texttt{max}}\}$ in CDM due to tidal stripping.

Cross Section — For this work, we explore a constant cross section per unit mass $\sigma_{\texttt{eff}}$ and run simulations with 5 cross-sections $\sigma_m/(\texttt{cm}^2/\texttt{g}) = 5, 10, 25$ and 50 and assume a SIDM model with elastic scattering. In this work we focus on core collapsed halos; strong lensing probes are most sensitive to core collapsed halos and insensitive to cored halos Gilman et al. [2021]. Constant SIDM cross sections constraints from clusters Eckert et al. [2022], Sagunski et al. [2021], Andrade et al. [2022] and dwarf and satellite galaxies Ando et al. [2025], Correa [2021] constrain constant cross section SIDM models to cross sections below $\sigma_m < 1\texttt{cm}^2/\texttt{g}$ which is lower cross section than considered in this work, and below the cross sections we find subhalos collapsing in our simulations. In our simulations, we do not consider SIDM particle interactions with the host; if SIDM interactions with host particles are not suppressed by (as is the case with a truly constant cross section SIDM models) subhalos are not expected to collapse due to evaporation Zeng et al. [2022]. Therefore, the constant cross sections in this work should be considered as an illustrative case study rather than something that is expected to be probed by strong lensing; mappings exist from velocity dependent cross sections to constant "effective" cross sections, which allow mapping our work onto velocity dependent cross section models, see References Yang and Yu [2022], Yang et al. [2023b], Outmezguine et al. [2023]. Finally, we note many velocity-dependent cross sections

considered in lensing analysis such as parametric cross section form considered in Gilman et al. [2021] asymptotes to a nearly constant cross section in the low velocity regime; these results can be viewed as exploring that regime: near constant cross section at low velocities and highly suppressed cross section on the scale of the host.

Core Collapse and Termination Conditions — Unlike field halos, the characteristic scale of subhalos may greatly decrease over time due to the effects of tidal stripping; to define core collapse we cannot simply choose a density threshold value at a fixed radius. Instead, we use the $R_{\texttt{max}}$ as the characteristic scale, where $R_{\texttt{max}}$ is the radius at which the maximum circular velocity $V_{\texttt{max}}$

$$V_{\texttt{max}} = \texttt{Max}\left\{\sqrt{\frac{GM(<r)}{r}}\right\} = \sqrt{\frac{GM(<R_{\texttt{max}})}{R_{\texttt{max}}}} \tag{3.7}$$

occurs. From Eq. 3.7, it follows that the average density within R_{max}, $\bar{\rho}_{\texttt{max}}\,(<R_{\texttt{max}})$ is

$$\bar{\rho}_{\texttt{max}}\,(<R_{\texttt{max}}) = \frac{3}{4\pi G}\left(\frac{V_{\texttt{max}}}{R_{\texttt{max}}}\right)^2 \tag{3.8}$$

. We define a subhalo as core collapsed and terminate with when the density within $R_{\texttt{max}}$ is

$$\bar{\rho}_{\texttt{max}}\,(<R_{\texttt{max}}) >= n\frac{3}{4\pi G}\left(\frac{\max\{V_{\texttt{max}}(t)\}}{\min\{R_{\texttt{max}}(t)\}}\right)^2 \tag{3.9}$$

. where $n > 1$ is an arbitrary factor that determines how deep into core collapse we resolve; for this work we chose $n = 10$.

3.3 Results

Here we discuss results from our suite of `T1DES` simulations. First, we compare results of subhalos with their isolated counterparts. We find that subhalos core collapse rates are strictly accelerated by the presense of a tidal field when compared to their isolated counterparts. Next, we compare our results to previous work including the Yang et al. [2024] parametric model for core collapse. We find good agreement with the parametric model and a similar order of magnitude core collapse acceleration when compared to Nadler et al. [2025b].

Core Collapse Acceleration — Figure 3.4 shows the the cross section per unit mass vs the core collapse fraction, the number of total core collapsed subhalos divided by total number of subhalos, $f_{\texttt{collapse}}$ for our T1DES simulations compared to the predictions from Yang et al. [2024] parametric model. We compare our tides subhalos run with a tidal field (T1DES SIDM stage) to their counterparts ran without a tidal field using the same initial conditions (T1DES isolated stage). We find that all isolated halos that collapse also collapse also collapse when run with a tidal field; the influence of a tidal field in our simulations strictly accelerates core collapse rate. However, we note that our results assume evaporation is negligible; for SIDM models where this is not the case, core collapse may delayed instead of accelerated. We find roughly similar acceleration rates to Nadler et al. [2025b] who compared the predicted core collapse rates for populations of isolated halos to subhalos in a SIDM model with a velocity dependent cross section; at $\sigma_m = 25\texttt{cm}^2/\texttt{g}$ nearly 4 times as many subhalos collapse when compared to their isolated counterparts and at $\sigma_m = 50\texttt{cm}^2/\texttt{g}$ (46%) nearly 2 times as many subhalos collapse. We find subhalo collapse for a cross section as low as $\sigma_{\texttt{m}} = 10\texttt{cm}^2/\texttt{g}$; at this cross section around 5% of the subhalos in our models collapse while none of their isolated counterparts do. For some SIDM models it may be possible that a non-negligible fraction of subhalos core collapse before any field halos do, due to accelerated core collapse from a tidal field.

Comparison with previous results — Figure 3.4 also shows the collapse fraction $f_{\texttt{collapse}}$, for our T1DES simulations compared to the predictions from the Yang et al. [2024] model; for this model, we show results obtained by integrating over the the tidal tracks from galacticus and from the T1DES CDM. We find our T1DES model is fairly consistent with the predictions from the Yang et al. [2024] model, though our T1DES model may predict slightly higher fractions of core collapsed halos. Interestingly our the Yang et al. [2024] predictions integrating over the galacticus tidal tracks appear to agree slightly better with our T1DES predictions than the the results obtained by integrating over the galacticus tidal tracks. We also note that the core collapse definition between our T1DES model and the Yang et al. [2024] model is different; Yang et al. [2024] defines core col-

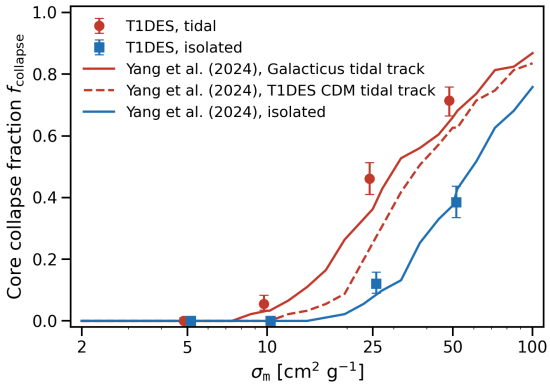


Figure 3.4: The cross section per unit mass vs the core collapse fraction for our `T1DES` simulations compared to as well as predictions from Yang et al. [2024]. In red: results from our t1des subhalo simulations (red dot with error bars), and the Yang models integrated over the the tidal tracks from `galacticus` (red solid) and from the `T1DES` CDM (red dotted) in blue. We find our `T1DES` model is fairly consistent with the predictions from the Yang et al. [2024] model, though our `T1DES` model may predict slightly higher fractions of core collapsed halos. In blue is the results for the isolated case, if the subhalos remained isolated, what fraction would collapse? Our `T1DES` model is consistent with the predictions of the Yang model here and predicts strictly lower collapse fractions for isolated halos. No halos collapse in the field that also do not collapse as a subhalo.

lapse as $\tau > 0.75$ while we use the $R_{\rm max}$ based density threshold given by Eq. 3.9. However, the difference in definitions likely only impacts the number of collapsed halos slightly; the final core collapse happens on short timescale compared to the entire SIDM evolution. We leave a more detailed comparison and investigating these possible discrepancies to future work.

3.4 Conclusion and Summary

In this work, we have presented `T1DES` , a novel method of simulating SIDM subhalos using a modified version of the `NSphere` 1D N-body code. Here we provide a summary of this work. In Section 3.2 we discuss the methods used in this paper

in three subsections. First, in Section 3.2.1 we discuss the `galacticus` model used to obtain the mass histories used by `T1DES` and the initial conditions used in our `T1DES` simulations. Then in Section 3.2.2 we discuss the dynamics implemented in our fork of `NSphere` ; in particular we focus on the tidal physics implemented in our simulations: a linear effective tidal field, calculated in a CDM first pass to match the mass history of a target subhalo. Thirdly, in Section 3.2.3 we give an overview of the SIDM physics implemented in NSphere and discuss our core collapse conditions for subhalos. a In the main section, Section 3.3 we discuss the main results from this work: with no evaporation, find that a tidal field strictly accelerates accelerates core collapse; and that this acceleration is cross section dependent and significant: up to a factor of nearly 4 time the number of subhalos collapse when compared to their isolated counterparts. Finally, here we discuss future work, discuss some potential caveats and provide concluding statements.

To correctly infer constraints on SIDM models through lensing studies, it will be important to thoroughly understand predicted core collapse rates. Our model gives a distinct advantage here: because it is much more computationally efficient than previous SIDM simulations, by making modifications to our pipeline it may now be practical to test a wider range of parameter and model space within SIDM. In addition to more thoroughly exploring the parameter space of specific SIDM due to the reduction of computational cost in our simulations By making some modifications to our code it is possible to explore a wide range of velocity and angle dependent cross sections. By adding in stochastic scattering chance of a particle interaction with the host simulating evaporation is possible within our model; we leave exploration of this to future work. Additionally, `NSphere` is not limited to dark matter only physics; it may be interesting to explore the effect of a central galaxy on core collapse times.

Our `T1DES` model is in reasonable agreement with `galacticus` the tidal tracks of `galacticus` halos, however some outlier halos disagree on order 50% of $V_{\texttt{max}}$; this disagreement is particularly pronounced for subhalos that are early in their evolution. This may be due to the initial conditions used by N-Sphere and the non physical stair stepping mass histories predicted by `galacticus` . It may be worth

further exploring seeding the `T1DES` pipeline with N-body mass histories rather than `galacticus` mass histories. We o not include any pre-infall SIDM evolution in our models; depending on the cross section, pre-infall SIDM evolution may be significant enough to form a core that effects the tidal physics of the subhalo. We leave investigating this regime to future work.

In conclusion, by simulating SIDM subhalo evolution in 1D, we are able to be much more computationally efficient in simulating SIDM subhalos when compared to traditional 3D N-body simulations by over an order of magnitude; our simulations are able to be simulated on a laptop and ran in the span of hours (this is greatly particle count and thus mass loss dependent). We shared Previous 1D SIDM subhalo simulations relied on coarse approximations of tidal stripping, such as approximating the tidal stripping as per orbit truncations to the mass profile; we include a much more realistic mass profile that reasonably approximates the mass history over time as well as the internal $R_{\texttt{max}}$ and $V_{\texttt{max}}$ properties of the subhalo. We find that under our assumption of no evaporation, subhalo core collapse is strictly accelerated when compared to isolated field halos. Additionally, we find that our simulations are in broad agreement with the Yang et al. [2024] parametric model.

Acknowledgments

Generative AI Disclosure — Generative AI (LLMs) were used in some portions of the research process for this work. Several Claude models were used as a part of this project: Claude Haiku, Sonnet, Opus and Fable[2] were used for coding assistance (including the generation of plots), summarization of background literature, as a general learning tool, transcriptions (taking an image of a diagram drawn by pencil and converting it to latex) and for several discussions about the project. We disclose that coding agents were used for several portions of this project such as writing portions of the tidal physics model, packaging `T1DES` into an ergonomic

[2]Many versions of Claude models were used throughout the project, starting at version 4.5 for haiku,sonnet and opus through to version 5.1 of fable

command line utility, implement various features into tides such as the "smoothed" and "isolated" modes, various assistance with analysis such as the generation of the figures in this work. Generative AI was also used for summarization of literature and as a general learning tool but was not used in the writing of this manuscript. For some early exploratory work, discussions and brainstorming with AI was tested "E.x What are some ideas for figures?". However, this usage was found to be infective and for this manuscript nearly all useful AI assistance was directed coding tasks, literature summarization, transcription and as and as a learning tool. We verify the output from AI coding agents in several ways: code generated by agents was reviewed, tests suggested by both agents and humans were set up and manually reviewed by the authors and finally, for a variety of tasks was duplicated and key results were checked by human made programs.

Chapter 4

Conclusion

In this work, we study the substructure of group mass lensing like halos and work towards developing informed priors of the populations of these subhalos in both CDM and alternate dark matter models. In Chapter 1 we gave a background of gravitational lensing probes, gave an overview of challenges studying dark matter subhalos and introduced our methods. Then in Chapter 2 we analyze state of the art `galacticus` simulations tuned to high resolution N-body simulations and develop a priors on the normalization of the subhalo mass function, it's scaling and estimate how this normalization is effected by tidal forces from a central galaxy. Next, in Chapter 3 we develop a novel, computationally efficient pipeline to develop priors core collapse rates of SIDM halos across a wide range of cross sections. Here, we summarize each of these chapters and provide final remarks.

The fundemental question we adress in Chapter 2 is "how can we develop informed priors on the normalization of the subhalo mass function for lensing subhalos to break degeneracies with the half mode mass"? With recent works have highlighted the potential for artifitial disruption in N-body artifictially deflating subhalo counts in N-body simulations, this question is particularly pressing. We start by running `galacticus` models with tidal physics tuned to ultra high resolution N-body simulations that are resistant to artificial disruption. Compare results from our `galacticus` model to a wide selection of N-body shows that our model agrees within a factor of ~ 2 with the N-body simulations. Additionally, motivated by the fact a central galaxy have been found to effect the subhalo normalization of

Milky Way like galaxies on the factor of ~ 2, we investigate a potential shortcomming of previous studies of lensing like group halos: the absensce of the potential of a central galaxy. We run a secondary suite of `galacticus` models including the potential of a realisticly evolving central galaxy and find that the potential of a central galaxy reduces the normalization on the 15% level, well within the systematic uncertainties of current lensing studies. Additionally, we recalibrate the projected redshift and halo mass scaling relations of the normalization of the projected subhalo mass function and compare it to predictions from analytic models. Finally, we suggest future anomalous flux ratio surveys incorporate a factor of ~ 2 uncertainty in their priors on the normalization of the subhalo mass function; we find that uncertainty in tidal stripping is the dominant uncertainty and that any effect from a central galaxy is negligible for current surveys.

Next, in Chapter 3 we develop a computationally efficient pipeline for simulating SIDM subhalos in the core collapse regime. We implement a linear tidal field in the 1-D N-body code `NSphere` and set it's strength by calculating the tidal field to match the mass loss rate of a target halo in a CDM first pass. In this way, we are able to realistically model tidal physics in 1D, greatly reducing the computational requirments of simulating SIDM subhalos enabling previosly infeasible studies of large numbers of subhalos. We run our pipeline for multiple realizations of `galacticus` halos and compare core collapse rates with analytic models; we find that core collapse of SIDM subhalos in the absences of evaporation is accelerated when compared to field halo counterparts; at some cross sections in our study nearly $4\times$ as many subhalos collapse when compared to field halo counterparts. In future works we plan to investigate velocity dependent SIDM cross sections as well as include subhalo evaporation.

In this work, we work to develop informative priors for subhalos in CDM as well as alternative dark matter models. We use and develop computationally efficient models able to sample the large volume of parameter space explored by parameter space. The informed priors we have developed in this work have been applied in lensing studies to break key degeneracies with observables. Additionally, as lensing studies move from the low data regime to having athoulsands of lensed

that are anticipated to be discovered by the Rubin, Euclid and Roman missions these priors will transition into testable predictions of the dark matter models themselves. Our work explores sets a benchmark for this new era of gravitational lensing constraints on dark matter: we explore systematic uncertanties in dark matter model predictions and lays out a new framework to computationally efficient exploration of the predictions of a large space of SIDM models.

Bibliography

Susmita Adhikari, Arka Banerjee, Kimberly K Boddy, Francis-Yan Cyr-Racine, Harry Desmond, Cora Dvorkin, Bhuvnesh Jain, Felix Kahlhoefer, Manoj Kaplinghat, Anna Nierenberg, et al. Astrophysical tests of dark matter self-interactions. *Reviews of Modern Physics*, 97(4):045004, 2025.

Shin'ichiro Ando, Kohei Hayashi, Shunichi Horigome, Masahiro Ibe, and Satoshi Shirai. Stringent constraints on self-interacting dark matter using milky-way satellite galaxies kinematics. *arXiv preprint arXiv:2503.13650*, 2025.

Kevin E Andrade, Jackson Fuson, Sophia Gad-Nasr, Demao Kong, Quinn Minor, M Grant Roberts, and Manoj Kaplinghat. A stringent upper limit on dark matter self-interaction cross-section from cluster strong lensing. *Monthly Notices of the Royal Astronomical Society*, 510(1):54–81, 2022.

Astropy Collaboration. Astropy: A community Python package for astronomy. *Astronomy & Astrophysics*, 558:A33, October 2013. doi: 10.1051/0004-6361/201322068.

Astropy Collaboration. The Astropy Project: Building an Open-science Project and Status of the v2.0 Core Package. *The Astronomical Journal*, 156(3):123, September 2018. doi: 10.3847/1538-3881/aabc4f.

Astropy Collaboration. The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package. *The Astrophysical Journal*, 935(2):167, August 2022. doi: 10.3847/1538-4357/ac7c74.

Nilanjan Banik, Gianfranco Bertone, Jo Bovy, and Nassim Bozorgnia. Probing the nature of dark matter particles with stellar streams. *Journal of Cosmology and Astroparticle Physics*, 2018(07):061, 2018.

Nilanjan Banik, Jo Bovy, Gianfranco Bertone, Denis Erkal, and TJL De Boer. Novel constraints on the particle nature of dark matter from stellar streams. *Journal of Cosmology and Astroparticle Physics*, 2021(10):043, 2021a.

Nilanjan Banik, Jo Bovy, Gianfranco Bertone, Denis Erkal, and T J L de Boer. Evidence of a population of dark subhaloes from Gaia and Pan-STARRS observations of the GD-1 stream. *Monthly Notices of the Royal Astronomical Society*, 502(2):2364–2380, 01 2021b. ISSN 0035-8711. doi: 10.1093/mnras/stab210. URL https://doi.org/10.1093/mnras/stab210.

Peter Behroozi, Risa H Wechsler, Andrew P Hearin, and Charlie Conroy. Universemachine: The correlation between galaxy growth and dark matter halo assembly from z=0-10. *Monthly Notices of the Royal Astronomical Society*, 488 (3):3143–3194, May 2019. ISSN 1365-2966. doi: 10.1093/mnras/stz1182. URL http://dx.doi.org/10.1093/mnras/stz1182.

Peter S Behroozi, Risa H Wechsler, and Hao-Yi Wu. The rockstar phase-space temporal halo finder and the velocity offsets of cluster cores. *The Astrophysical Journal*, 762(2):109, 2012.

Peter S Behroozi, Risa H Wechsler, Yu Lu, Oliver Hahn, Michael T Busha, Anatoly Klypin, and Joel R Primack. Mergers and mass accretion for infalling halos both end well outside cluster virial radii. *The Astrophysical Journal*, 787(2):156, 2014.

Andrew J. Benson. Galacticus: A semi-analytic model of galaxy formation. *New Astronomy*, 17(2):175–197, 2012. ISSN 1384-1076. doi: https://doi.org/10.1016/j.newast.2011.07.004. URL https://www.sciencedirect.com/science/article/pii/S1384107611000807.

Andrew J. Benson and Xiaolong Du. Tidal tracks and artificial disruption of cold

dark matter haloes. *Monthly Notices of the Royal Astronomical Society*, 517(1): 1398–1406, November 2022. doi: 10.1093/mnras/stac2750.

Ana Bonaca, David W Hogg, Adrian M Price-Whelan, and Charlie Conroy. The spur and the gap in gd-1: Dynamical evidence for a dark substructure in the milky way halo. *The Astrophysical Journal*, 880(1):38, 2019.

JR Bond, Shaun Cole, George Efstathiou, and Nick Kaiser. Excursion set mass functions for hierarchical gaussian fluctuations. *Astrophysical Journal, Part 1 (ISSN 0004-637X), vol. 379, Oct. 1, 1991, p. 440-460. Research supported by NSERC, NASA, and University of California.*, 379:440–460, 1991.

Jo Bovy, Denis Erkal, and Jason L Sanders. Linear perturbation theory for tidal streams and the small-scale cdm power spectrum. *Monthly Notices of the Royal Astronomical Society*, 466(1):628–668, 2017.

Richard G Bower. The evolution of groups of galaxies in the press–schechter formalism. *Monthly Notices of the Royal Astronomical Society*, 248(2):332–352, 1991.

Michael Boylan-Kolchin, Chung-Pei Ma, and Eliot Quataert. Dynamical friction and galaxy merging time-scales. *Monthly Notices of the Royal Astronomical Society*, 383(1):93–101, 2008.

Subrahmanyan Chandrasekhar. Dynamical friction. i. general considerations: the coefficient of dynamical friction. *Astrophysical Journal*, 97:255–262, 1943.

Shaun Cole, Cedric G. Lacey, Carlton M. Baugh, and Carlos S. Frenk. Hierarchical galaxy formation. *Monthly Notices of the Royal Astronomical Society*, 319(1): 168–204, 11 2000a. ISSN 0035-8711. doi: 10.1046/j.1365-8711.2000.03879.x. URL `https://doi.org/10.1046/j.1365-8711.2000.03879.x`.

Shaun Cole, Cedric G. Lacey, Carlton M. Baugh, and Carlos S. Frenk. Hierarchical galaxy formation. *Monthly Notices of the Royal Astronomical Society*, 319(1): 168–204, 11 2000b. ISSN 0035-8711. doi: 10.1046/j.1365-8711.2000.03879.x. URL `https://doi.org/10.1046/j.1365-8711.2000.03879.x`.

Camila A Correa. Constraining velocity-dependent self-interacting dark matter with the milky way's dwarf spheroidal galaxies. *Monthly Notices of the Royal Astronomical Society*, 503(1):920–937, 2021.

WJG De Blok, Fabian Walter, Elias Brinks, C Trachternach, SH Oh, and Robert C Kennicutt. High-resolution rotation curves and galaxy mass models from things. *The Astronomical Journal*, 136(6):2648, 2008.

Giulia Despali, Simona Vegetti, Simon D M White, Carlo Giocoli, and Frank C van den Bosch. Modelling the line-of-sight contribution in substructure lensing. *Monthly Notices of the Royal Astronomical Society*, 475(4):5424–5442, 01 2018. ISSN 0035-8711. doi: 10.1093/mnras/sty159. URL `https://doi.org/10.1093/mnras/stky159`.

Benedikt Diemer. Colossus: A python toolkit for cosmology, large-scale structure, and dark matter halos. *The Astrophysical Journal Supplement Series*, 239(2): 35, 2018.

Benedikt Diemer and Michael Joyce. An accurate physical model for halo concentrations. *The Astrophysical Journal*, 871(2):168, 2019.

Veronica Dike, Daniel Gilman, and Tommaso Treu. Strong lensing constraints on primordial black holes as a dark matter candidate. *Monthly Notices of the Royal Astronomical Society*, 522(4):5434–5441, 2023.

Scott Dodelson and Lawrence M Widrow. Sterile neutrinos as dark matter. *Physical Review Letters*, 72(1):17, 1994.

Elena D'Onghia, Volker Springel, Lars Hernquist, and Dusan Keres. Substructure depletion in the milky way halo by the disk. *The Astrophysical Journal*, 709(2): 1138, 2010.

Gregory A Dooley, Annika HG Peter, Mark Vogelsberger, Jesús Zavala, and Anna Frebel. Enhanced tidal stripping of satellites in the galactic halo from dark matter self-interactions. *Monthly Notices of the Royal Astronomical Society*, 461(1):710–727, 2016.

Xiaolong Du, Andrew Benson, Zhichao Carton Zeng, Tommaso Treu, Annika H. G. Peter, Charlie Mace, Fangzhou Jiang, Shengqi Yang, Charles Gannon, Daniel Gilman, Anna. M. Nierenberg, and Ethan O. Nadler. Tidal evolution of cored and cuspy dark matter halos. *arXiv e-prints*, art. arXiv:2403.09597, March 2024. doi: 10.48550/arXiv.2403.09597.

Xiaolong Du, Andrew Benson, Zhichao Carton Zeng, Tommaso Treu, Annika HG Peter, Charlie Mace, Fangzhou Jiang, Shengqi Yang, Charles Gannon, Daniel Gilman, et al. Tidal evolution of cored and cuspy dark matter halos. *Physical Review D*, 110(2):023019, 2024.

Xiaolong Du, Daniel Gilman, Tommaso Treu, Andrew Benson, and Charles Gannon. Faster than sam: An empirical model for the tidal evolution of dark matter substructure around strong gravitational lenses. *arXiv preprint arXiv:2503.07728*, 2025.

Toshihiro Ebisu, Tomoaki Ishiyama, and Kohei Hayashi. Constraining self-interacting dark matter with dwarf spheroidal galaxies and high-resolution cosmological n-body simulations. *Physical Review D*, 105(2):023016, 2022.

D Eckert, STEFANO Ettori, A Robertson, R Massey, E Pointecouteau, D Harvey, and IG McCarthy. Constraints on dark matter self-interaction from the internal density profiles of x-cop galaxy clusters. *Astronomy & Astrophysics*, 666:A41, 2022.

Raphaël Errani and Julio F Navarro. The asymptotic tidal remnants of cold dark matter subhaloes. *Monthly Notices of the Royal Astronomical Society*, 505(1): 18–32, 2021.

Raphaël Errani and Jorge Peñarrubia. Can tides disrupt cold dark matter subhaloes? *Monthly Notices of the Royal Astronomical Society*, 491(4):4591–4601, February 2020. doi: 10.1093/mnras/stz3349.

Raphaël Errani and Jorge Peñarrubia. Can tides disrupt cold dark matter subhaloes? *Monthly Notices of the Royal Astronomical Society*, 491(4):

4591–4601, 12 2019. ISSN 0035-8711. doi: 10.1093/mnras/stz3349. URL https://doi.org/10.1093/mnras/stz3349.

Davide Fiacconi, Piero Madau, Doug Potter, and Joachim Stadel. Cold Dark Matter Substructures in Early-type Galaxy Halos. *The Astrophysical Journal*, 824(2):144, June 2016. doi: 10.3847/0004-637X/824/2/144.

Charles Gannon, Anna Nierenberg, Andrew Benson, Ryan Keeley, Xiaolong Du, and Daniel Gilman. Dark matter substructure: A lensing perspective. *Physical Review D*, 112(2):023532, 2025. doi: 10.1103/kk2n-q4ps.

L. Gao, S. D. M. White, A. Jenkins, F. Stoehr, and V. Springel. The subhalo populations of ΛCDM dark haloes. *Monthly Notices of the Royal Astronomical Society*, 355(3):819–834, 12 2004. ISSN 0035-8711. doi: 10.1111/j.1365-2966.2004.08360.x. URL https://doi.org/10.1111/j.1365-2966.2004.08360.x.

L. Gao, J. F. Navarro, C. S. Frenk, A. Jenkins, V. Springel, and S. D. M. White. The Phoenix Project: the dark side of rich Galaxy clusters. *Monthly Notices of the Royal Astronomical Society*, 425(3):2169–2186, 09 2012. ISSN 0035-8711. doi: 10.1111/j.1365-2966.2012.21564.x. URL https://doi.org/10.1111/j.1365-2966.2012.21564.x.

Daniel Gilman, Simon Birrer, Anna Nierenberg, Tommaso Treu, Xiaolong Du, and Andrew Benson. Warm dark matter chills out: constraints on the halo mass function and the free-streaming length of dark matter with eight quadruple-image strong gravitational lenses. *Monthly Notices of the Royal Astronomical Society*, 491(4):6077–6101, 12 2019a. ISSN 0035-8711. doi: 10.1093/mnras/stz3480. URL https://doi.org/10.1093/mnras/stz3480.

Daniel Gilman, Simon Birrer, Tommaso Treu, Anna Nierenberg, and Andrew Benson. Probing dark matter structure down to 107 solar masses: flux ratio statistics in gravitational lenses with line-of-sight haloes. *Monthly Notices of the Royal Astronomical Society*, 487(4):5721–5738, 2019b.

Daniel Gilman, Jo Bovy, Tommaso Treu, Anna Nierenberg, Simon Birrer, Andrew Benson, and Omid Sameie. Strong lensing signatures of self-interacting dark matter in low-mass haloes. *Monthly Notices of the Royal Astronomical Society*, 507(2):2432–2447, 2021.

Daniel Gilman, Andrew Benson, Jo Bovy, Simon Birrer, Tommaso Treu, and Anna Nierenberg. The primordial matter power spectrum on sub-galactic scales. *Monthly Notices of the Royal Astronomical Society*, 512(3):3163–3188, 2022.

Daniel Gilman, Yi-Ming Zhong, and Jo Bovy. Constraining resonant dark matter self-interactions with strong gravitational lenses. *Physical Review D*, 107(10): 103008, 2023.

Daniel Gilman, Simon Birrer, Anna Nierenberg, and Maverick SH Oh. Turbocharging constraints on dark matter substructure through a synthesis of strong lensing flux ratios and extended lensed arcs. *Monthly Notices of the Royal Astronomical Society*, 533(2):1687–1713, 2024.

Oleg Y Gnedin, Lars Hernquist, and Jeremiah P Ostriker. Tidal shocking by extended mass distributions. *The Astrophysical Journal*, 514(1):109, 1999.

Oleg Y Gnedin, Andrey V Kravtsov, Anatoly A Klypin, and Daisuke Nagai. Response of dark matter halos to condensation of baryons: cosmological simulations and improved adiabatic contraction model. *The Astrophysical Journal*, 616(1): 16, 2004.

Sheridan B Green and Frank C van den Bosch. The tidal evolution of dark matter substructure – I. subhalo density profiles. *Monthly Notices of the Royal Astronomical Society*, 490(2):2091–2101, 10 2019. ISSN 0035-8711. doi: 10.1093/mnras/stz2767. URL `https://doi.org/10.1093/mnras/stz2767`.

Brendan F Griffen, Alexander P Ji, Gregory A Dooley, Facundo A Gómez, Mark Vogelsberger, Brian W O'Shea, and Anna Frebel. The caterpillar project: a large suite of milky way sized halos. *The Astrophysical Journal*, 818(1):10, 2016.

Jiaxin Han, Shaun Cole, Carlos S. Frenk, and Yipeng Jing. A unified model for the spatial and mass distribution of subhaloes. *Monthly Notices of the Royal Astronomical Society*, 457(2):1208–1223, 02 2016. ISSN 0035-8711. doi: 10.1093/mnras/stv2900. URL https://doi.org/10.1093/mnras/stv2900.

Qiuhan He, Andrew Robertson, James W Nightingale, Aristeidis Amvrosiadis, Shaun Cole, Carlos S Frenk, Samuel C Lange, Shubo Li, Ran Li, Xiaoyue Cao, et al. Not so dark, not so dense: an alternative explanation for the lensing subhalo in sdss j0946+ 1006. *The Astrophysical Journal Letters*, 991(2):L53, 2025.

Lars Hernquist. An analytical model for spherical galaxies and bulges. *Astrophysical Journal, Part 1 (ISSN 0004-637X), vol. 356, June 20, 1990, p. 359-364.*, 356:359–364, 1990.

Lilian Jiang, Shaun Cole, Till Sawala, and Carlos S Frenk. Orbital parameters of infalling satellite haloes in the hierarchical λcdm model. *Monthly Notices of the Royal Astronomical Society*, 448(2):1674–1686, 2015.

Felix Kahlhoefer, Manoj Kaplinghat, Tracy R Slatyer, and Chih-Liang Wu. Diversity in density profiles of self-interacting dark matter satellite halos. *Journal of Cosmology and Astroparticle Physics*, 2019(12):010–010, 2019.

Marc Kamionkowski and Kris Sigurdson. Gravothermal collapse of self-interacting dark-matter halos with anisotropic velocity distributions. *arXiv preprint arXiv:2510.23705*, 2025.

Marc Kamionkowski and Kris Sigurdson. Evolution of self-gravitating spherical dark-matter halos with and without new physics. *Physical Review D*, 113(10): 103026, 2026.

Marc Kamionkowski, Kris Sigurdson, and Oren Slone. Numerical evolution of self-gravitating halos of self-interacting dark matter. *Physical Review Letters*, 136 (20):201001, 2026.

Ryan E Keeley, Anna M Nierenberg, Daniel Gilman, Charles Gannon, Simon Birrer, Tommaso Treu, Andrew J Benson, Xiaolong Du, KN Abazajian, T Anguita, et al. Jwst lensed quasar dark matter survey ii: Strongest gravitational lensing limit on the dark matter free streaming length to date. *arXiv preprint arXiv:2405.01620*, 2024.

David Klemmer, Moritz S Fischer, Kimberly K Boddy, Manoj Kaplinghat, and Laura Sagunski. Survival of the most compact: the life and death of satellite halos in self-interacting dark matter. *arXiv preprint arXiv:2603.19362*, 2026.

Aaron D Ludlow, Sownak Bose, Raúl E Angulo, Lan Wang, Wojciech A Hellwing, Julio F Navarro, Shaun Cole, and Carlos S Frenk. The mass–concentration–redshift relation of cold and warm dark matter haloes. *Monthly Notices of the Royal Astronomical Society*, 460(2):1214–1232, 2016.

Philip Mansfield, Elise Darragh-Ford, Yunchong Wang, Ethan O Nadler, Benedikt Diemer, and Risa H Wechsler. Symfind: Addressing the fragility of subhalo finders and revealing the durability of subhalos. *The Astrophysical Journal*, 970 (2):178, 2024.

Shude Mao and Peter Schneider. Evidence for substructure in lens galaxies? *Monthly Notices of the Royal Astronomical Society*, 295(3):587–594, 1998.

Maxim Markevitch, Anthony H Gonzalez, D Clowe, Alexey Vikhlinin, W Forman, Ch Jones, S Murray, and W Tucker. Direct constraints on the dark matter self-interaction cross section from the merging galaxy cluster 1e 0657–56. *The Astrophysical Journal*, 606(2):819–824, 2004.

Quinn Minor, Sophia Gad-Nasr, Manoj Kaplinghat, and Simona Vegetti. An unexpected high concentration for the dark substructure in the gravitational lens sdssj0946+ 1006. *Monthly Notices of the Royal Astronomical Society*, 507(2): 1662–1683, 2021.

Ethan O Nadler, Arka Banerjee, Susmita Adhikari, Yao-Yuan Mao, and Risa H

Wechsler. Signatures of velocity-dependent dark matter self-interactions in milky way-mass halos. *The Astrophysical Journal*, 896(2):112, 2020.

Ethan O Nadler, Simon Birrer, Daniel Gilman, Risa H Wechsler, Xiaolong Du, Andrew Benson, Anna M Nierenberg, and Tommaso Treu. Dark matter constraints from a unified analysis of strong gravitational lenses and milky way satellite galaxies. *The Astrophysical Journal*, 917(1):7, 2021.

Ethan O Nadler, Philip Mansfield, Yunchong Wang, Xiaolong Du, Susmita Adhikari, Arka Banerjee, Andrew Benson, Elise Darragh-Ford, Yao-Yuan Mao, Sebastian Wagner-Carena, et al. Symphony: Cosmological zoom-in simulation suites over four decades of host halo mass. *The Astrophysical Journal*, 945(2): 159, 2023.

Ethan O Nadler, Rui An, Daneng Yang, Hai-Bo Yu, Andrew Benson, and Vera Gluscevic. Cozmic. iii. cosmological zoom-in simulations of self-interacting dark matter with suppressed initial conditions. *The Astrophysical Journal*, 986(2): 129, 2025a.

Ethan O Nadler, Demao Kong, Daneng Yang, and Hai-Bo Yu. Sidm concerto: compilation and data release of self-interacting dark matter zoom-in simulations. *The Astrophysical Journal*, 991(1):69, 2025b.

Hiroya Nishikawa, Kimberly K Boddy, and Manoj Kaplinghat. Accelerated core collapse in tidally stripped self-interacting dark matter halos. *Physical Review D*, 101(6):063009, 2020.

Nadav Joseph Outmezguine, Kimberly K Boddy, Sophia Gad-Nasr, Manoj Kaplinghat, and Laura Sagunski. Universal gravothermal evolution of isolated self-interacting dark matter halos for velocity-dependent cross-sections. *Monthly Notices of the Royal Astronomical Society*, 523(3):4786–4800, 2023.

Hannah Parkinson, Shaun Cole, and John Helly. Generating dark matter halo merger trees. *Monthly Notices of the Royal Astronomical Society*, 383(2):557–564, 2008.

Planck Collaboration. Planck2018 results: Vi. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, September 2020a. ISSN 1432-0746. doi: 10.1051/0004-6361/201833910. URL http://dx.doi.org/10.1051/0004-6361/201833910.

Planck Collaboration. Planck2018 results: Vi. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, September 2020b. ISSN 1432-0746. doi: 10.1051/0004-6361/201833910. URL http://dx.doi.org/10.1051/0004-6361/201833910.

Devon M Powell, Simona Vegetti, JP McKean, Simon DM White, Elisa GM Ferreira, Simon May, and Cristiana Spingola. A lensed radio jet at milli-arcsecond resolution–ii. constraints on fuzzy dark matter from an extended gravitational arc. *Monthly Notices of the Royal Astronomical Society: Letters*, 524(1):L84–L88, 2023.

William H Press and Paul Schechter. Formation of galaxies and clusters of galaxies by self-similar gravitational condensation. *Astrophysical Journal, Vol. 187, pp. 425-438 (1974)*, 187:425–438, 1974.

Scott W Randall, Maxim Markevitch, Douglas Clowe, Anthony H Gonzalez, and Marusa Bradač. Constraints on the self-interaction cross section of dark matter from numerical simulations of the merging galaxy cluster 1e 0657–56. *The Astrophysical Journal*, 679(2):1173–1180, 2008.

Miguel Rocha, Annika HG Peter, James S Bullock, Manoj Kaplinghat, Shea Garrison-Kimmel, Jose Onorbe, and Leonidas A Moustakas. Cosmological simulations with self-interacting dark matter–i. constant-density cores and substructure. *Monthly Notices of the Royal Astronomical Society*, 430(1):81–104, 2013.

Laura Sagunski, Sophia Gad-Nasr, Brian Colquhoun, Andrew Robertson, and Sean Tulin. Velocity-dependent self-interacting dark matter from groups and clusters of galaxies. *Journal of Cosmology and Astroparticle Physics*, 2021(01):024–024, 2021.

Neev Shah and Susmita Adhikari. The abundance of core-collapsed subhaloes in sidm: insights from structure formation in λcdm. *Monthly Notices of the Royal Astronomical Society*, 529(4):4611–4623, 2024.

Shiyin Shen, H. J. Mo, Simon D. M. White, Michael R. Blanton, Guinevere Kauffmann, Wolfgang Voges, J. Brinkmann, and Istvan Csabai. The size distribution of galaxies in the Sloan Digital Sky Survey. *Monthly Notices of the Royal Astronomical Society*, 343(3):978–994, August 2003. doi: 10.1046/j.1365-8711.2003.06740.x.

Ravi K Sheth and Giuseppe Tormen. Large-scale bias and the peak background split. *Monthly Notices of the Royal Astronomical Society*, 308(1):119–126, 1999.

Yiping Shu, Joel R Brownstein, Adam S Bolton, Léon VE Koopmans, Tommaso Treu, Antonio D Montero-Dorta, Matthew W Auger, Oliver Czoske, Raphaël Gavazzi, Philip J Marshall, et al. The sloan lens acs survey. xiii. discovery of 40 new galaxy-scale strong lenses. *The Astrophysical Journal*, 851(1):48, 2017.

V. Springel, J. Wang, M. Vogelsberger, A. Ludlow, A. Jenkins, A. Helmi, J. F. Navarro, C. S. Frenk, and S. D. M. White. The Aquarius Project: the subhaloes of galactic haloes. *Monthly Notices of the Royal Astronomical Society*, 391(4): 1685–1711, 12 2008. ISSN 0035-8711. doi: 10.1111/j.1365-2966.2008.14066.x. URL `https://doi.org/10.1111/j.1365-2966.2008.14066.x`.

Maryam Tajalli, Simona Vegetti, Conor M O'Riordan, Simon DM White, Christopher D Fassnacht, Devon M Powell, John P McKean, and Giulia Despali. Sharp–ix. the dense, low-mass perturbers in b1938+ 666 and j0946+ 1006: implications for cold and self-interacting dark matter. *Monthly Notices of the Royal Astronomical Society*, 543(1):540–564, 2025.

Sean Tulin and Hai-Bo Yu. Dark matter self-interactions and small scale structure. *Physics Reports*, 730:1–57, 2018.

Hannah C Turner, Mark R Lovell, Jesús Zavala, and Mark Vogelsberger. The onset of gravothermal core collapse in velocity-dependent self-interacting dark

matter subhaloes. *Monthly Notices of the Royal Astronomical Society*, 505(4): 5327–5339, 2021.

Frank C van den Bosch and Go Ogiya. Dark matter substructure in numerical simulations: a tale of discreteness noise, runaway instabilities, and artificial disruption. *Monthly Notices of the Royal Astronomical Society*, 475 (3):4066–4087, 01 2018. ISSN 0035-8711. doi: 10.1093/mnras/sty084. URL https://doi.org/10.1093/mnras/sty084.

Frank C. Van den Bosch, Giuseppe Tormen, and Carlo Giocoli. The mass function and average mass-loss rate of dark matter subhaloes. *Monthly Notices of the Royal Astronomical Society*, 359(3):1029–1040, 05 2005. ISSN 0035-8711. doi: 10.1111/j.1365-2966.2005.08964.x. URL https://doi.org/10.1111/j.1365-2966.2005.08964.x.

Frank C van den Bosch, Go Ogiya, Oliver Hahn, and Andreas Burkert. Disruption of dark matter substructure: fact or fiction? *Monthly Notices of the Royal Astronomical Society*, 474(3):3043–3066, 11 2017. ISSN 0035-8711. doi: 10.1093/mnras/stx2956. URL https://doi.org/10.1093/mnras/stx2956.

S Vegetti, LVE Koopmans, A Bolton, T Treu, and R Gavazzi. Detection of a dark substructure through gravitational imaging. *Monthly Notices of the Royal Astronomical Society*, 408(4):1969–1981, 2010.

S Vegetti, G Despali, MR Lovell, and W Enzi. Constraining sterile neutrino cosmologies with strong gravitational lensing observations at redshift z 0.2. *Monthly Notices of the Royal Astronomical Society*, 481(3):3661–3669, 2018.

Mark Vogelsberger, Jesus Zavala, and Abraham Loeb. Subhaloes in self-interacting galactic dark matter haloes. *Monthly Notices of the Royal Astronomical Society*, 423(4):3740–3752, 2012.

Jeremy J Webb and Jo Bovy. High-resolution simulations of dark matter subhalo disruption in a milky-way-like tidal field. *Monthly Notices of the Royal Astronomical Society*, 499(1):116–128, 2020.

David H Weinberg, James S Bullock, Fabio Governato, Rachel Kuzio de Naray, and Annika HG Peter. Cold dark matter: controversies on small scales. *Proceedings of the National Academy of Sciences*, 112(40):12249–12255, 2015.

Simon DM White and Carlos S Frenk. Galaxy formation through hierarchical clustering. *Astrophysical Journal, Part 1 (ISSN 0004-637X), vol. 379, Sept. 20, 1991, p. 52-79. Research supported by NASA, NSF, and SERC.*, 379:52–79, 1991.

Simon DM White and Martin J Rees. Core condensation in heavy halos: a two-stage theory for galaxy formation and clustering. *Monthly Notices of the Royal Astronomical Society*, 183(3):341–358, 1978.

Dandan Xu, Dominique Sluse, Liang Gao, Jie Wang, Carlos Frenk, Shude Mao, Peter Schneider, and Volker Springel. How well can cold dark matter substructures account for the observed radio flux-ratio anomalies. *Monthly Notices of the Royal Astronomical Society*, 447(4):3189–3206, 01 2015. ISSN 0035-8711. doi: 10.1093/mnras/stu2673. URL `https://doi.org/10.1093/mnras/stu2673`.

Daneng Yang and Hai-Bo Yu. Gravothermal evolution of dark matter halos with differential elastic scattering. *Journal of Cosmology and Astroparticle Physics*, 2022(09):077, 2022.

Daneng Yang, Ethan O Nadler, and Hai-Bo Yu. Strong dark matter self-interactions diversify halo populations within and surrounding the milky way. *The Astrophysical Journal*, 949(2):67, 2023a.

Daneng Yang, Ethan O Nadler, Hai-Bo Yu, and Yi-Ming Zhong. A parametric model for self-interacting dark matter halos. *Journal of Cosmology and Astroparticle Physics*, 2024(02):032, 2024.

Shengqi Yang, Xiaolong Du, Andrew J Benson, Anthony R Pullen, and Annika H G Peter. A new calibration method of sub-halo orbital evolution for semi-analytic models. *Monthly Notices of the Royal Astronomical Society*, 498(3):

3902–3913, 09 2020. ISSN 0035-8711. doi: 10.1093/mnras/staa2496. URL https://doi.org/10.1093/mnras/staa2496.

Shengqi Yang, Xiaolong Du, Zhichao Carton Zeng, Andrew Benson, Fangzhou Jiang, Ethan O Nadler, and Annika HG Peter. Gravothermal solutions of sidm halos: Mapping from constant to velocity-dependent cross section. *The Astrophysical Journal*, 946(1):47, 2023b.

Donald G York, J Adelman, John E Anderson Jr, Scott F Anderson, James Annis, Neta A Bahcall, JA Bakken, Robert Barkhouser, Steven Bastian, Eileen Berman, et al. The sloan digital sky survey: Technical summary. *The Astronomical Journal*, 120(3):1579, 2000.

Zhichao Carton Zeng, Annika HG Peter, Xiaolong Du, Andrew Benson, Stacy Kim, Fangzhou Jiang, Francis-Yan Cyr-Racine, and Mark Vogelsberger. Core-collapse, evaporation, and tidal effects: the life story of a self-interacting dark matter subhalo. *Monthly Notices of the Royal Astronomical Society*, 513(4): 4845–4868, 2022.

Zhichao Carton Zeng, Annika HG Peter, Xiaolong Du, Shengqi Yang, Andrew Benson, Francis-Yan Cyr-Racine, Fangzhou Jiang, Charlie Mace, and R Benton Metcalf. Till the core collapses: The evolution and properties of self-interacting dark matter subhalos. *arXiv preprint arXiv:2310.09910*, 2023.

Andrew R Zentner, Andreas A Berlind, James S Bullock, Andrey V Kravtsov, and Risa H Wechsler. The physics of galaxy clustering. i. a model for subhalo populations. *The Astrophysical Journal*, 624(2):505, 2005.

Xingyu Zhang, Hai-Bo Yu, Daneng Yang, and Ethan O Nadler. The gd-1 stellar stream perturber as a core-collapsed self-interacting dark matter halo. *The Astrophysical Journal Letters*, 978(2):L23, 2025.